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A maturity model for assessing the implementation of Industry 5.0 in manufacturing SMEs: learning from theory and practice

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ABSTRACT

Digitalization is facing the transition from Industry 4.0 to Industry 5.0, enlarging the technocentric vision of technologies to embrace sustainable, resilience and human-centered design objectives. Although aware of the benefits of digitalization, many companies have not yet undertaken this process or are only at the beginning stages, seeking support and expertise. Firstly, it is important to understand the company's current technological level and, therefore, establish an improvement strategy. Over time, several authors have made efforts to develop maturity models capable of assessing a company's Industry 4.0 technological level. However, given the recent shift toward Industry 5.0, it is necessary to understand how such tools can effectively address the three pillars of Industry 5.0 when evaluating digital maturity. This study contributes to the literature on Industry 5.0 by proposing a tool to assess the maturity of companies in Industry 5.0 practices, focusing on sustainability, resilience and human centricity in manufacturing processes. This Industry 5.0 Maturity Model consists of a self-assessment questionnaire, designed to assess the current maturity level of the company across seven maturity dimensions (double assessment Industry 5.0 integrated and pillar-based), and a roadmap that outlines improvement strategies and operational information to enhance the company' maturity across these dimensions. The proposed maturity model is applied in a multiple case study to collect empirical evidence from four Italian SMEs operating in the manufacturing industry. The results show a good Industry 5.0 maturity, with differences across the various dimensions influenced by the characteristics of the analyzed sectors and products. The low level of embedded technologies leaves room for future improvements in products and services, aligned with sustainability, resilience and human-centric perspectives. Theoretical and practical implications of the study are also discussed.

1. Introduction

Digital technologies are capable of influencing businesses and society by driving changes in processes, business models and organizational structures (Ulas, 2019; Verhoef et al., 2021). Since 2011, the Industry 4.0 paradigm has supported companies, particularly in the manufacturing (Rojko, 2017), on their digital transformation journey, providing a set of technologies (the nine pillars) that improve production efficiency and decision-making through data integration and visualization. As a result of the crisis condition generated by the COVID-19 pandemic, it emerged that digitalized companies were capable of reacting better to the difficulties (Almeida et al., 2020; Amankwah-Amoah et al., 2021) because more agile in processes reconfiguration (Staufen AG, 2020). This is even more relevant for small and medium-sized enterprises (SMEs), which in many countries represent the largest part of the economic system. For example, the share of SMES in 2019 was 94.8 % in Italy (ISTAT, 2022) and 99.4 % in Germany (Statistisches Bundesamt., 2019). This trend is confirmed across Europe: 99 % of European businesses are SMEs, providing jobs to >85 million European citizens.[1] Digitalization can help SMEs compete with larger businesses by improving the quality of products and services, reducing cost (Ramdani et al., 2022), increasing communication and collaboration efficiency, and opening new business areas (Mittelstand Digital, 2019; Telekom Deutschland, 2022). For this reason, it is crucial that these businesses undergo a digital transformation.

In the current scenario, digitalization is facing the transition from the paradigm of Industry 4.0 to that of Industry 5.0, shifting from the technocentric vision to a more synergistic one. Leveraging on the relevance of sustainability issues in the current worldwide scenario, Industry 5.0 was defined as an industry model capable of achieving "societal goals beyond jobs and growth to become a provider of prosperity, by making production respect the boundaries of our planet and

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[1] https://single-market-economy.ec.europa.eu/smes_en#:~:text=What%20the%20European%20Commission%20does,heart%20of%20innovation%20and%20entrepreneurship.

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placing the wellbeing of the industry worker at the center of the production process" (European Commission, 2020). Therefore, this new technological paradigm was conceived around three main pillars: sustainability, resilience and human-centered design (more details are in Section 2.1).

Although there is a strong awareness of the benefits derived from digitalization, it is clear that many companies have not yet undertaken this process or are only at the beginning stages (Fromhold-Eisebith et al., 2021; StaufenAG, 2020), seeking support and expertise. This is especially true for SMEs, which face numerous challenges and risks during a digitalization process (Pelletier and Cloutier, 2019; Wiesner et al., 2018), such as the need for investment (Wan and Leirmo, 2023), expertise development (Han and Trimi, 2022), and issues related to information security and data protection (Almeida et al., 2020).

To guide a company through a digitalization process, it is first important to understand its current technological state by evaluating its so called "maturity level"(Cozmiuc and Pettinger, 2021). For this purpose, over time, maturity models (MMs) have been developed to assess the initial state of a company and provide suggestions for improving its digital maturity (Kırmızı and Kocaoglu, 2022). Over time, several authors have worked on developing MMs capable of assessing a company's I4.0 technological level. However, given the recent shift from the I4.0 paradigm to I5.0, it is necessary to understand how such tools can cover the three pillars of I5.0 when assessing digital maturity. Due to the lack of currently developed MMs for I5.0 (more details in Section 2.2), and building on key insights from I4.0 MMs, this study proposes a MM to assess the capability of companies to adopt I4.0 technologies while moving toward the implementation of the three I5.0 pillars: sustainability, resilience and human-centric approach. The proposed I5.0 MM is conceived as a prescriptive model; therefore, it was composed by a tool for the maturity assessment (self-assessment questionnaire) and a roadmap for the maturity improvement (guidelines for improvement). The I5.0 MM is composed by 7 maturity dimensions (adapted from Wagire et al. (2021)) and 38 maturity items conceived to assess, in a double layer of analysis, the integrated I5.0 maturity level and the pillar-based maturity level across sustainability, resilience and human-centred design. It was applied in a multiple case study in order to collect evidence from 4 Italian SMEs operating in manufacturing industry and specifically in agri-food and fashion sectors. During this application phase, empirical evidence about the current MM level of the selected companies (in its integrated and pillar-based perspectives) and the guidelines for improvement were discussed. My results indicate a good level of maturity in I5.0, with differences across the various dimensions driven by the characteristics of the analyzed sectors and products. The low level of embedded technologies highlights opportunities for companies to improve products and services in line with sustainability, resilience, and human-centric perspectives.

This study contributes to the I5.0 literature by proposing a tool to assess the maturity of companies in adopting I5.0 practices, with a focus on sustainability, resilience, and human-centricity of manufacturing processes. The added value lies in the empirical results obtained from the multiple case study, including a comparison of four companies from two distinct manufacturing sectors, as well as the discussed implications for research and practices, which suggest direction for future research.

2. Literature review

2.1. The transition from Industry 4.0 to Industry 5.0

The rapid advancement of information technologies is ushering in the Fourth Industrial Revolution, placing the world on the brink of transformative change (Skobelev and Borovik, 2017). The phenomenon of I4.0 has introduced a range of cutting-edge technologies, organized into nine core pillars,[2] which have the potential to fundamentally reshape the design and production processes of goods and services (Daim and Faili, 2019; Narula et al., 2021). This shift has enabled industries to improve the operational efficiency of factories and plants enhancing existing technological frameworks with new tools for data integration, visualization, and decision-making support. These innovations are paving the way for a transformative approach to manufacturing and service delivery. The technology-driven focus of I4.0 emphasizes improving efficiency and flexibility in industry, often overlooking industrial sustainability and worker welfare (Huang et al., 2022). However, the current worldwide scenario requires companies to take sustainability issues into account in their business activities: respect natural environment, preserve raw materials, safe biodiversity, reduce pollutants and waste, ensuring social progress and equity, protect territorial economies (Bai and Sarkis, 2023; Osmanoglu et al., 2020; Rueda et al., 2017; Schoenherr and Talluri, 2013) are just some of the needs required by governments and consumers. Addressing these challenges requires to companies to pay more attention to environmental and social sustainability issues adopting new human and environmental centered strategies for industrial modernization—placing people and societal needs at the forefront. In this perspective, I5.0 complements the existing I4.0 paradigm by encouraging research and innovation toward a sustainable, resilient and human-centric industry transition (Xu et al., 2021). The Fifth Industrial Revolution leverages on I4.0 technologies proposing the evolution of the paradigm toward a new vision for an innovative, resilient, and socially conscious industry that not only respects planetary boundaries but also minimizes environmental impact (Müller, 2020). In 2021, the European Commission urged European industries to reassess their societal roles, emphasizing the importance of fostering long-term prosperity (European Commission. Directorate General for Research and Innovation., 2021). Accordingly, I5.0 aims to harness the power of industry to achieve broader societal goals, moving beyond mere growth to serve as a driver of resilience, green transition, and social-economic well-being (Saniuk et al., 2022). The vision for I5.0 is a profound societal transformation, building upon the shift toward a "smart society" (Gerlitz, 2015) already initiated by I4.0 technologies. I5.0 takes this concept further, evolving the "smart society" into a "super-smart society," where scientific and technological advancements guide the creation of sustainable, resilient and human-centered world—referred to as "Society 5.0" (S5.0) (Carayannis et al., 2023; Huang et al., 2022). Therefore, sustainability, resilience and human-centered represent the three main pillars that encourage companies to adopt the I5.0 vision, supporting the transition of the entire societal context toward the S5.0 model. Below is a brief description of these pillars.

2.1.1. Sustainability

Sustainability refers to the need to reduce the environmental impact of production activities (He et al., 2024). A key challenge in achieving this goal is the efficient utilization of natural resources, whether as energy or materials (Caggiano et al., 2023). Technologies can influence the sustainability pillar positively or negatively by increasing network reliability and resilience in distributed energy resource (Newcomb et al., 2013), increasing or reducing energy consumption (Oláh et al., 2020), providing data analysis insights to minimize energy consumption (Rocha et al., 2021) or production waste (Wang et al., 2024), implementing circular economy approaches to reduce waste and pollution (Ivanov, 2023), and producing energy from cleaned and renewable sources (Holroyd, 2022).

[2] Big Data, Cybersecurity, Additive Manufacturing, Horizontal and Vertical Systems Integration, Augmented Reality, Simulation, Cloud Computing, Internet of Things and Autonomous Robots.

2.1.2. Resilience

Resilience is defined as a "measure of the persistence of systems and their ability to absorb changes and perturbations and maintain the same relationships among populations or state variables" (Holling, 1973). In manufacturing, it represents the capability of the production system to withstand disruptions and catastrophic events (Van Oudenhoven et al., 2023). According to Caggiano et al. (2023), the introduction of I5.0 technologies in production systems can negatively impact system resilience, requiring companies to make additional efforts to address technical disruption caused by malfunctions or cyber-attacks. This challenge can be mitigated through modular design strategies that enhance flexibility and simplicity in production systems. The interaction between the human workforce and smart machines can help companies create robust and resilient production systems (Leng et al., 2022). The adoption of predictive analytics, artificial intelligence, and digital twins can support effective decision-making in the face of unforeseen events (Nassif et al., 2010; Semeraro et al., 2023). Cloud computing and data analytics ensure remote work capabilities and business continuity (Gupta et al., 2023), while blockchain technologies enhance traceability, flexibility and adaptability, binging resilience benefits to production systems and value chains (Leng et al., 2022).

2.1.3. Human-centered design

Human-centered design is problem-solving approach used to develops solutions to problems (e.g., product, process, service, system) by involving human perspective in each phase of the solution development. It enhances effectiveness and efficiency guarantying, at the same time, the human well-being and satisfaction.[3] In the evolutionary digitalization path, following the introduction of I4.0 robots in production activities, I5.0 envisions a symbiotic relationship between employees and collaborative robots (Rožanec et al., 2022) which are designed to increase the production quantity and/or quality but also to assure employees' safety and, more generally, to give an answer to the human needs improving working conditions (Caggiano et al., 2023). This approach leads also to customized products and services capable of better satisfying the consumers' needs (Shifrin and Michel, 2022) and to implement feedback mechanism to discover, understand and then satisfy all customer needs (Or et al., 2023). The integration of IoT into production plants and value chains has enabled the collection of more data, not only about production activities but also about operators' task. This necessitates that companies update their procedures to safeguard employees' privacy and protect sensitive data (Caggiano et al., 2023).

The goal of I5.0 in implementing a S5.0 is of ensuring that all citizens enjoy high-quality, comfortable, and fulfilling lives by providing necessary goods and services at the right time, achieved through the integration of cyberspace and physical space powered by I4.0 technologies (Huang et al., 2022). This evolution is neither simple nor straightforward. It requires the development of strategies to guide companies toward long-lasting digital transformation, providing support in measuring sustainability, resilience and human-centered design in technology innovation, while also fostering further advancements.

2.2. Review of existing Industry 4.0 and Industry 5.0 maturity models

MMs are tools used to assess the maturity level of a company in a specific domain or to measure the level of implementation of a process in relation to future goal (Gökalp et al., 2017). They are based on the premise that people, organizations, functional areas and processes evolve over time, progressing through a development process toward higher levels of maturity (Mittal et al., 2018). Each level of a MM represents a starting point from which evolution toward higher levels can be planned and implemented. Maturity dimensions represent capabilities within a domain of interest, are measurable by levels, and can be identified through the application of various quantitative methods such as literature reviews, case studies, focus groups (O'Donovan et al., 2016).

A MM can be conceived for: i) descriptive purposes, to assess the current state (as-is situation) of an organization or process; ii) prescriptive purposes, to indicate how to approach maturity level improvement with the goal of positively impacting business value, such as by developing a maturity improvement roadmap; and iii) comparative purposes, to enable cross-benchmarking, allowing comparisons of similar practices across various organizations, with the aim of developing maturity comparisons across industries (Asdecker and Felch, 2018; García-Mireles et al., 2012; Maier et al., 2011; Röglinger et al., 2012).

Several authors have been involved in developing MMs for I4.0, while the field of I5.0 MMs is still in its early stage. By applying the following search query on the Scopus database, a corpus of documents was retrieved: ((“Industry 4.0” OR “Industry 5.0) AND “Maturity Model”). After selecting empirical studies published in relevant journals over the past five years (2019–2024) and considering MMs used in their literature reviews, 18 MMs for I4.0 were identified (more details are provided in the Supplementary Material). In the analysis sample, only three studies focused on I5.0 MMs. However, these are literature reviews that summarize I4.0 MMs without proposing empirical evidences from the development of new MMs for I5.0 (Caggiano et al., 2023; Hein-Pensel et al., 2023; Zizic et al., 2022). The selected I4.0 MMs were analyzed according the following elements: the purpose and type of the MM, the methodology followed in the MM development, dimensions (and items) of the MM, maturity levels, tool for assessment, and the presence of a roadmap. A synthesis of this analysis is provided in the Supplementary Material of this study. The analysis reveals the presence of MMs focused on measuring I4.0 maturity in the specific case of SMEs (Simetinger and Basl, 2022; Spaltini et al., 2022; Zoubek et al., 2021). However, not all of these models were developed according to a well-established methodology. Dimensions and items differ from model to model, although a general tendency emerges to consider both organizational aspects (e.g., people, change, culture, strategy) and technological aspects (e.g., smart devices, smart factories, smart products and services). Additionally, there is a range of maturity levels between four and seven, with a greater use of questionnaires as a data collection tool (Amaral and Peças, 2021; Ávila Bohórquez and Gil Herrera, 2022; Caiado et al., 2021; Çınar et al., 2021; Colli et al., 2018; Rauch et al., 2020; Santos and Martinho, 2020; A. Schumacher et al., 2016; A. Schumacher et al., 2019; Senna et al., 2023; Simetinger and Basl, 2022; Ünal et al., 2022; Ünal et al., 2022; Wagire et al., 2021; Zoubek et al., 2021). There is also a limited presence of tools to support sustained improvement (Caiado et al., 2021; Rauch et al., 2020; Spaltini et al., 2022).

3. A research methodology for the development of the maturity model

Over time, several authors have provided guidelines on how to develop a MM (e.g., Hevner et al. (2004) and Becker et al. (2009) and De Bruin et al. (2005)). Among them, I chose to adopt the methodology proposed by De Bruin et al. (2005), as it is well-established and widely applied for the development of MMs across various knowledge domains (Santos and Martinho, 2020), such as industrial analytics (O'Donovan et al., 2016), human-centered business process (Mamoghli et al., 2018), and delivery processes in supply chain (Asdecker and Felch, 2018). This methodology consists of six interactive stages, where results from one stage may require revisiting and revising previous stages.

The first stage, *'Defining the purpose of the model'*, aims to clarify the scope within which the model will be applied. The focus of the model is established in this stage, determining whether it is general or specific to a particular field, and identifying the target audience (e.g., academics, managers, government institutions or their combination). The aim,

[3] ISO 9241-210:2019

focus and target of the proposed model are discussed in Section 4.1.

The second stage, '*Defining the model design*', aims of understanding the needs of the target audience. Several questions can guide this phase, such as: *why use this MM? How can this MM be applied in different organizational structures? Who should be involved in applying this MM? What can be achieved by applying this MM?* A preliminary analysis was conducted by interviewing 20 managers from Italian manufacturing SMEs, with the aim of understanding the current adoption level of technological assets and the companies' awareness of the benefits of I4.0. The results are discussed in Section 4.2. This analysis led me to choose the development of a prescriptive model composed of a self-assessment questionnaire and a roadmap.

The third stage, '*Establishment and validation of the model*', aims of establishing and validating the model by defining its contents, what elements to measure, and how. Generally, this phase begins with a literature review to identify all the contents useful for model establishment (Santos and Martinho, 2020). Given that the I5.0 paradigm is the natural and current evolution of the I4.0 paradigm, it is appropriate to start from the already established I4.0 MMs in order to leverage the research efforts previously conducted in the scientific community. As reported in Section 2.2, the literature on MMs was explored to collect various MMs related to I4.0 and I5.0, identifying 18 I4.0 MMs and none for I5.0. The literature review was conducted according to the guidelines provided by Torraco (2005), following a concept-centric perspective (Webster and Watson, 2002). Therefore, the identified MMs were thoroughly analyzed, categorized and compared according to the following criteria: characteristics of the model (aim and type of the model), the development methodology, maturity dimensions and items, the number of maturity levels and the tools used (questionnaire, interview, the roadmap). The details of the research protocol and the literature synthesis were provided as supplementary material. Among the I4.0 MMs analyzed, the one proposed by Wagire et al. (2021) represents a very robust model, built from an extensive and recent analysis of the literature and validated through case studies. Therefore, the following specific reasons contributed to choosing the MM proposed by Wagire et al. (2021) as a good starting point for the establishment of the proposed I5.0 model:

- The model was established following a well-established research methodology (Becker et al., 2009; Hevner et al., 2004), including several phases such as: literature review and MMs comparison, expert interviews, conceptual design, model testing, and validation in a case study.
- The model's dimensions are fully representative of the entire manufacturing context, considering aspects such as people and culture, awareness of I4.0, organizational strategy, value chain and processes, technologies for smart manufacturing, technologies for products and services, and I4.0 technologies. Moreover, these dimensions were established following a careful analysis of the literature.
- The items used to describe the dimensions were well-defined and clearly described. These were defined triangulating the results from the literature, consulting experts in the manufacturing sector, and conducting questionnaire-based interviews with directors of institutes operating in the field of I4.0.
- The mathematical model used to assess the company's maturity level (Fuzzy Analytic Hierarchy Process (FAHP)), based on the use of weights for dimensions and items, is useful for enhancing the model's ability to assess the maturity levels in specific context (e.g., sectors, company size), allowing for customization in the case of manufacturing SMEs.

Starting from this model represents a careful and conscious choice, driven by the desire to use an existing, valid, and validated model. Indeed, conducting a new literature analysis aimed at defining the dimensions and items representative of I.5 cannot be done without considering the results relating to I4.0, since I5.0 is a progressive, evolutionary and emerging context. Therefore, this analysis would lead to identification of the same topics present in Wagire et al. (2021), with the difference that they should be explored through the lens of I5.0. For this reason, this study begins with the I4.0 MM items proposed by Wagire et al. (2021), reinterpreting them from the new perspective introduced by I5.0: "sustainable, resilient and human-centered" technology innovation. In this way, this study represents the opportunity to conduct a longitudinal analysis of the I4.0-I5.0 evolution, starting from the constructs formulated in 2021 by Wagire et al., and evolving them from the viewpoint of the I5.0 key topics. This choice leads to outlining a new and interesting scenario useful for both academic and industrial contexts, from which important considerations arise. The procedure of starting from the comparison analysis of existing MMs, selections valuable dimensions and items from them, and adapting these for a new purpose (or gap filling) represents a well-established practice already followed by several authors in the definition of their MMs (e.g., Colli et al., 2018; Santos and Martinho, 2020; Senna et al., 2023; Spaltini et al., 2022; Wagire et al., 2021).

Further information on the structure of the model (dimensions and items), the mathematical model, and the tools for application are provided in Sections 3.1 and 3.2.

The fourth stage, '*Pilot test of the model*', aims of validating the model in terms of content and form, in order to evaluate the clarity of the contents, as well as the practicality and usability of the model. This can be done through the selection of a series of companies for conducting the pilot application, either from the same sector or from different sectors, in line with the purpose of the proposed model. This phase was implemented through a multiple case study involving four SMEs from the agri-food and fashion sectors (Section 3.3).

The fifth stage, '*Distribution of the model*', involves the release of the model for wide-scale use. To make the model accessible and widely accepted, two important issues must be addressed. In this stage, there is the opportunity to identify organizations that can benefit from the use of the proposed model, validating its application across more organizations. This is useful for standardizing the model and promoting its global acceptance. The sixth phase, '*Model maintenance*', depends on the resources needed to update and use the model over time. These resources, such as a data repository, are needed to support the use of the model by a large volume of users. For example, if the model becomes available online, several resources are needed to continuously update its interface, keep up with technological developments, and maintain its contents. The last two stages are not the focus of this study and represent follow-ups to the current work.

Fig. 1 summarizes the first four stages followed for the implementation of the proposed I5.0 MM.

3.1. Development of I5.0 maturity model and supporting tools

Starting from the I4.0 maturity dimensions proposed by Wagire et al. (2021), the structure of the proposed I5.0 MM was developed defining dimensions, items, maturity levels. Specifically, the maturity dimensions were inherited as-is and slightly redefined in terms, leading to the consideration of seven dimensions useful for assessing the maturity level of manufacturing SMEs in a digitalization process based on I5.0 principles. Leveraging a literature review and considering the specific aim of the model, several items were established for each dimension, resulting in a total of 38 maturity items. These items were designed to assess the digitalization maturity along the three pillars of I5.0: sustainability, resilience and human-centered design. A focus group of five researchers in the field of digital transformation was established with the aim of assessing the capability of each maturity item in contributing to the building of the three I5.0 pillars. From this discussion, it emerged that some maturity items were capable of simultaneously addressing the three I5.0 pillars, some addressed only two, and others addressed just one pillar (see the last column in Table 3). The formulation of statements

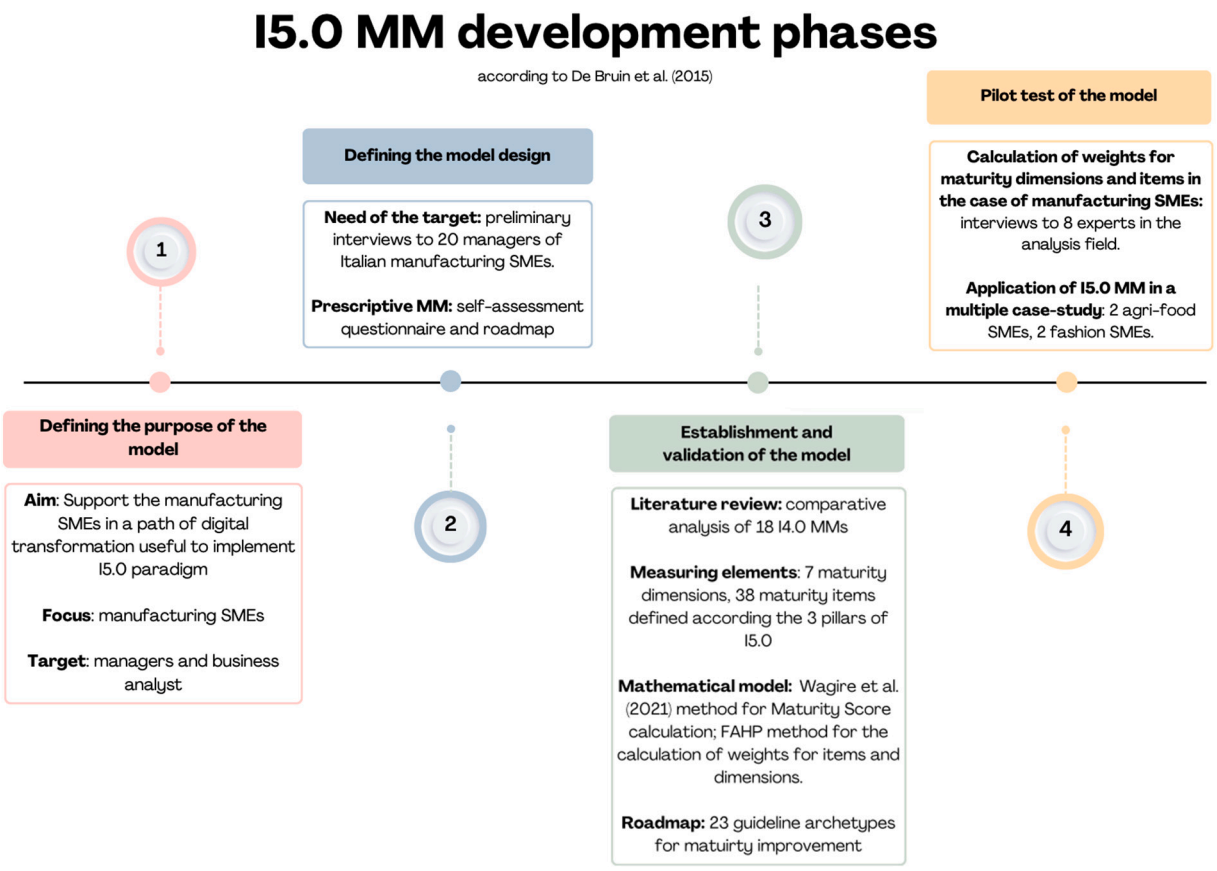


Fig. 1. - Research methodology (adapted from De Bruin et al. 2015).

useful for measuring the multi-pillar items was extensively discussed during the focus group. If an item contributes more than one pillar, should the statement to be duplicate or triplicate, or should inclusive statement be proposed? The focus group experts expressed concerns about conducting the analysis in an overly analytical way, as it would have been challenging to convey the subtle nuances among the three I5.0 pillars to the respondent - replicating the same statement for all three pillars. Therefore, for multi-pillar items, they suggested creating statements that perform an initial integrated assessment of the company's maturity level in "sustainable, resilient and human centered production". Once the maturity value of each item across the entire company has been measured, a further phase of analysis can then be conducted to investigate how this integrated maturity value of I5.0 can be broken down across the three I5.0 pillars. The item maturity score, calculated for each item in integrated manner, is shown to the respondent, who is then asked what percentage of the value can be attributed to the achievement of sustainability, resilience, and human-centered design objectives. The answers collected during this second interview allow for the creation of a second level of knowledge, in which each I5.0 pillar is assessed according to the related maturity items. A company can choose to implement or not this in-depth analysis, or alternatively, it can choose to investigate a single pillar from the beginning, selecting the relevant items and appropriately modifying the statements.

Since this model was born as a prescriptive model, it is composed of a questionnaire for self-assessment and a set of guidelines that support managers and business analysts in developing appropriate suggestions to guide the company toward an I5.0 maturity improvement path (roadmap). The questionnaire consist of 4 sections.

A descriptive section and a glossary precede the questionnaire, to promote understanding of the questions and ensure correct investigation. The second section is a general and introductory section composed by 7 multiple-choice questions aimed at collecting information about: the legal form of the company, number of employees, average turnover value, the profile of the interviewee, the interviewee's perception of the company's technological level compared to the sector average, the interviewee's perception of the company's I5.0 pillars levels (sustainability, resilience and human-centered design) compared to the sector average, and the interviewee's perception of the company's use of technology according to the I5.0 perspective. The third section was focused on performing the integrative assessment of I5.0, measuring the maturity levels of dimensions and items in sustainable, resilience and human-centered production. Therefore, the descriptive statements of items were used to ask to the interviewee to assess the I5.0 maturity level of the company with respect to the statement, using a 5-point Likert scale where "1" indicates "No or low maturity" and "5" indicates "A high level of maturity".

The final section was focused on performing the I5.0 pillar-based assessment, measuring the maturity levels of dimensions and items for each I5.0 pillar separately. Here, open descriptive statements were used, which the respondent could fill in providing their opinion.

The entire questionnaire was subjected to a content validation, which involved three researchers in digital development field. The discussion in this focus group, led to improvements the formulation of the statements and the addition of missing terms in the glossary of the introductory section.

After data collection, following the procedure summarized in Fig. 2 and using an ad-hoc designed spreadsheet, the answers obtained were used to calculate, first the maturity score for items and dimensions and, second, the overall I5.0 maturity score of the company according to the integrated assessment procedure. The I5.0 pillars maturity score of dimensions and items was then calculated according the I5.0 pillar-based assessment procedure.

As described in Fig. 2, the Overall Maturity Score represents the weighted average of all dimension maturity scores in the company and,

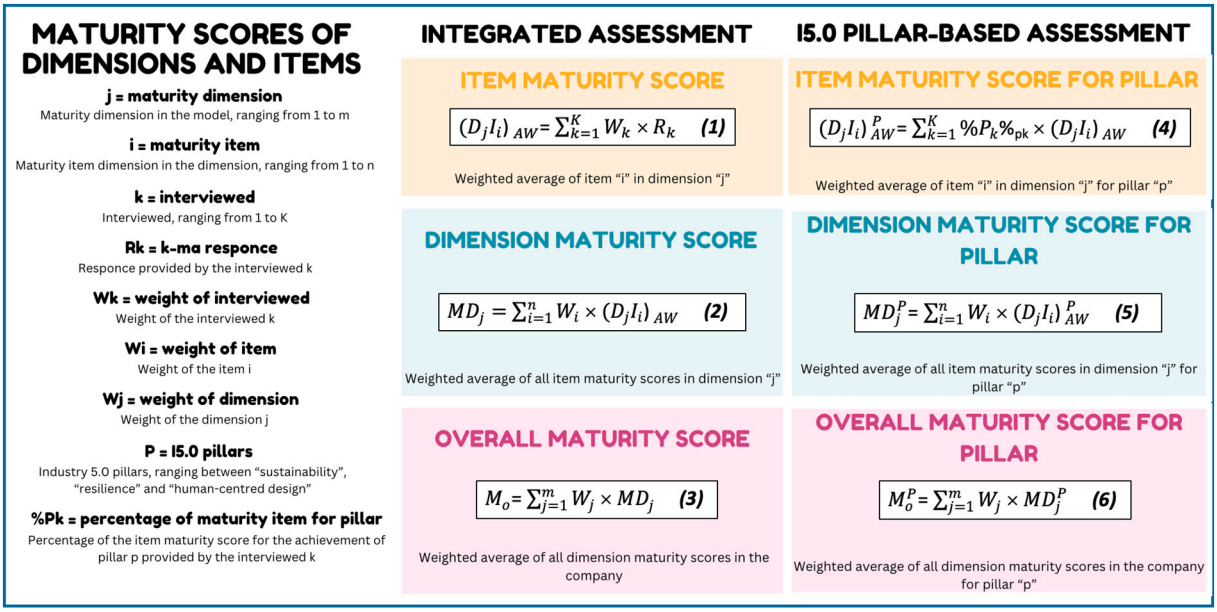


Fig. 2. - Mathematical model for calculation of I5.0 and I5.0 pillars maturity scores of dimensions and items (adapted from Wagire et al., 2021).

therefore, it is calculated as the sum of the product of the Dimension Maturity Score and the relative weight, both in the case of I5.0 integrated and I5.0 pillar-based assessment. Finally, the obtained maturity scores are translated into maturity levels according to the conversion described in Table 1.

3.2. *Defining the weights for maturity dimensions and items: The case of manufacturing SMEs*

It is well established that every maturity item has a different significance level in the maturing process of the company (Keskin et al., 2022; Lichtblau et al., 2015; D. M. Schumacher, 2016; Wagire et al., 2021). For example, the adoption of "Additive manufacturing and 3D printing" may contribute differently to I5.0 maturity than the adoption of "Blockchain Technology". Moreover, it is certainly not obvious that the level of contribution of each item and a dimension is the same regardless of the industrial sector or the size of the company. For example, there are structural differences that lead small and medium-sized enterprises to have different levels of significance from large enterprises. Therefore, to identify the contribution of each maturity item in determining the overall maturity level of the company, the multi-criteria decision-making technique FAHP is applied. The FAHP method incorporates Fuzzy theory into the Analytical Hierarchy Process (AHP) (Saaty, 2008), which is a decision-making tool widely used in the solution of multi-criteria decision-making problems starting from the pairwise comparison between different alternatives. Since the AHP does not take into account the fact that a judgment can be vague, it has been improved by the logic brought by the Fuzzy approach. In the FAHP methodology, the comparison between pairs of alternatives is performed using linguistic variables represented by triangular numbers (Kilincci and Onal, 2011).

Table 1
- Maturity level for Industry 5.0 and Industry 5.0 pillars (adapted from Wagire et al., 2021).

Overall Maturity Score - M_o Overall Maturity Score for PILLAR - M_o^p	Maturity level
$4.00 \leq M_o \leq 5.00$	Level 4: Senior
$3.00 \leq M_o \leq 4.00$	Level 3: Advanced
$2.00 \leq M_o \leq 3.00$	Level 2: Beginner
$1.00 \leq M_o \leq 2.00$	Level 1: Outsider

Specifically, as part of this study, Buckley's method (Buckley, 1985) was used to determine the weights of dimensions and items. The procedure is summarized in Table 2 and can be useful for replicating the study in other industrial sectors by submitting the questionnaire to respective experts.

Fig. 3 shows an example of the comparison matrix used in the stage 1 for data collection. Specifically, the questionnaire, which is composed of several comparison matrices for dimensions and items within each dimension, was validated using a focus group consisting of three researchers in digital development. The aim was of improving the formulation of the statement and establish the glossary of terms that supported the questionnaire. Afterward, it was submitted to eight experts in the field: three professors and 1 researcher from three different universities, each an expert in innovation management, technologies for sustainable production, digital transformation, and sustainable oriented innovation; and four managers of small and medium-sized manufacturing companies involved in digital transformation initiatives. This choice was made to ensure a comprehensive and complementary perspective between the academic and the industrial sectors, with the goal of obtaining objective values for the weights of dimensions and items, making the model more effective for SMEs in the manufacturing sector. The weights calculated according to the procedure described here are reported in Table 3.

3.3. *Testing and implementation phase: A multiple case study*

The testing and validation phase was conducted using the multiple case study technique. The case study provides a clear answer to the "how" and "why" of specific phenomena occurring in a given context. Its purpose is to accurately understand the case under consideration. Therefore, a case study can be defined as an empirical investigation that examines a contemporary phenomenon within its real-life context, particularly when the boundaries between the phenomenon and the contexts are not clearly evident. The case study investigation technically addresses the particular situation in which there are many more variables of interest than data problems. Consequently, it makes use of multiple sources of evidence, with necessary triangulation of data, and benefits from the development of the preceding theoretical propositions to guide data collection and analysis (Yin Robert, 2005). The multiple case study offers advantages over the single case study because it increase the reliability of the collected data by adding rigor to the study of

Table 2
– Procedure to calculate weights for dimensions and items (adapted from Buckley, 1985).

Step 1: comparison among dimensions and item	Creation of a comparison matrix for dimensions and a comparison matrix for items in each dimension. The comparison matrices ask the decision maker to select the level of importance of one dimension compared to another according to a Saaty scale (equally important, not very important, quite important, strongly important, absolutely important). Translation of Saaty scale in the Fuzzy triangular scale (respectively (1,1,1); (2,3,4); (4,5,6); (6,7,8); (9,9,9)). Following this, the Fuzzy comparison matrix is created, where $d^k ij$ indicates the preference of the Kth decision maker in the comparison between dimensions/items i^{th} and j^{th}, expressed according to the Fuzzy triangular numbers. $\widetilde{A}^k = \begin{bmatrix} \widetilde{d}^k_{11} & \widetilde{d}^k_{12} & \ldots & \widetilde{d}^k_{1n} \\ \widetilde{d}^k_{21} & \ldots & \ldots & \widetilde{d}^k_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ \widetilde{d}^k_{n1} & \widetilde{d}^k_{n2} & \ldots & \widetilde{d}^k_{nn} \end{bmatrix}$
Step 2: Calculation of the average of the preferences of the various decision makers	If there are multiple decision makers, the average of the preferences from the various decision makers is calculated, according to the following equation: $\widetilde{d}_{ij} = \frac{\sum_{k=1}^{K} \widetilde{d}^k_{ij}}{K}$
Step 3: Calculation of the pair wise contribution matrix	Using the logic expressed in the previous phases, the average of the preferences from the various decision makers, is used to calculate the comparison matrix for the dimensions or for the items within a dimension as follows: $\tilde{A} = \begin{bmatrix} \widetilde{d_{11}} & \cdots & \widetilde{d_{1n}} \\ \vdots & \ddots & \vdots \\ \widetilde{d_{n1}} & \cdots & \widetilde{d_{nn}} \end{bmatrix}$
Step 4: Calculation of the triangular value	The geometric mean of the Fuzzy comparison values for each dimension or item is calculated according to the following equation, which represents the triangular Fuzzy value, where n is the number of dimensions or items within the dimension: $\widetilde{r}_i = \left(\prod_{j=1}^{n} \widetilde{d}_{ij} \right)^{1/n}, \quad i=1, 2,\ldots,n$
Step 5: Fuzzy weight calculation	To calculate the Fuzzy weight for each dimension or each item within the dimension, follows these steps in succession: - Calculate the sum of the Fuzzy comparison values. - Compute its inverse value. - Finally, the inverse value is sorted in an ascending order to construct the sorted Fuzzy triangular number. The triangular weight is then calculated by multiplying the Fuzzy triangular number r of the dimension or item by the ordered Fuzzy triangular number: $w_i = (x_i * a), (y_i * b), (z_i * c) = (\alpha, \beta, \delta)$ It is still a fuzzy triangular number that can be transformed into an absolute value using mathematical averaging: $M_i = \frac{(\alpha + \beta + \delta)}{3}$

the observed phenomenon. In addition, replicating the study across multiple contexts allows for a broader comparison and analysis of the results. The selection of cases is a crucial step, and their choice can be guided by various factors. In this study, I chose to analyze four different business case studies pertaining to two main sectors: agri-food and fashion. The choice of these sectors was based on the following reasons. It is widely known that the manufacturing sector has a significant impact on sustainability (for example CO_2 emissions, use of natural resources, workplace safety, and community protection) (Lena et al., 2022; Mangla et al., 2024; Yang et al., 2023). Among the various manufacturing sectors, agri-food and fashion were considered with the greatest sustainability impact from production activities (Abbate et al., 2023; Dangelico et al., 2022; Provin et al., 2021). Moreover, these sectors are emblematic of Made in Italy products worldwide. Given the reveled sustainability impacts, manufacturing companies are adopting sustainability strategies to improve their business quality. According to an ISTAT survey conducted in April 2023,[4] in 2022, 59,5 % of Italian manufacturing companies took sustainability actions. Among them, 50,3 % implemented actions for environmental protection, 44,6 % focused on social sustainability (human-centered), and 36,8 % on economic sustainability. The Italian agri-food and fashion sectors (textiles, clothing, and footwear) are moderately engaged in such sustainability practices, with sustainability actions reported by 75 % of agri-food companies and 65 % of fashion companies. Finally, the ISTAT survey revealed that large companies are, on average, the most active in all sustainability practices: >4/5 of large companies (81,5 %) take sustainability actions, compared to only 36,1 % of small companies. These factors make Italian SMEs belonging to agri-food and fashion industries eligible for this study.

The selection of individual case studies in the two sectors was guided by a desire to analyze companies with a high level of technology adoption. The goal was to understand whether these technologies enabled them to mature along a sustainable production path or if they merely used to support business efficiency within their technocentric view. A summary of the companies selected as case studies, along with their rationale, is shown in Fig. 4. For each case study, the self-assessment questionnaire was administered to three different employees with diverse profiles (e.g., marketing manager, sales specialist, production manager) to ensure a diversity of opinions during the analysis phase. This approach allowed for the integrated assessment of I5.0. After calculating the item maturity scores, dimension maturity scores and the overall maturity score, these were discussed with the respondents to gather data for the I5.0 pillar-based assessment and obtain valuable information for design the roadmap for improvement.

4. The proposed Industry 5.0 maturity model

4.1. Scope of the maturity model

The scope of the proposed I5.0 MM is to support the manufacturing SMEs in their digital transformation journey, facilitating the implementation of the I5.0 paradigm. The focus is on the capability of manufacturing SMEs to use I4.0 technologies to create sustainable, resilient and human-centric products and production systems. This is achieved by leveraging product and process information, which can be better managed using I4.0 technologies to assess, monitor, and improve company performance across the three I5.0 pillars. In line with this scope, the proposed model is a prescriptive model. Once the current level of I5.0 maturity and its associated pillars are assessed, the model provides a roadmap to guide the company in establishing a set of guidelines aimed at improving the maturity level across the different dimensions. Therefore, the target audience for the model includes managers of manufacturing SMEs and business analyst.

[4] https://www.istat.it/comunicato-stampa/pratiche-sostenibili-nelle-imprese-nel-2022-e-nel-2023-2025/

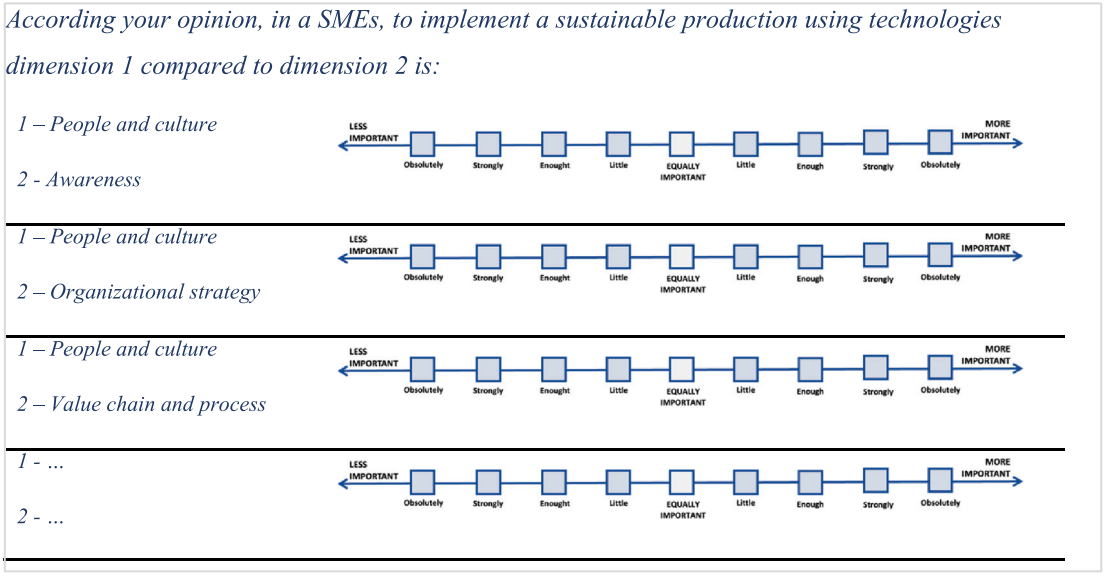


Fig. 3. - Example of comparison matrix for data collection.

4.2. Design of the maturity model

In order to understand the needs of manufacturing SMEs, a preliminary analysis was conducted by interviewing 20 managers of Italian manufacturing SMEs. The aim of the interview was to assess the level of adoption of technological assets and the awareness of the benefits that I4.0 can bring to the company. From this preliminary analysis, it emerged that half of the sample had not yet undertaken a digitalization journey, although there was a good level of awareness regarding the benefits these technologies can offer. A difficulty in adopting these technologies with a systems-oriented approach was also identified. Therefore, the proposed model can be used by manufacturing SMEs to assess their current maturity level in the implementation of I5.0 as a whole and across its three pillars. The MM for I5.0 consist of an assessment questionnaire (to be use in two subsequent phases of data collection) and a roadmap to support future improvement. These tools were initially designed for general purpose, but are customized for manufacturing SMEs through the FAHP method, as explained in Section 3.2.

4.3. Maturity model building and validation

The proposed MM includes 38 maturity measurement items organized in 7 dimensions. Table 3 describes maturity dimensions and items, their relation to the I5.0 pillars, and shows the weight calculated for manufacturing SMEs according to the procedure described in Section 3.2. As shown in Table 2, all maturity dimensions contribute to the implementation of the three pillars of I5.0. This is especially true for dimensions 1 and 2, where all maturity items contribute to building the three pillars. In the other dimensions, some maturity items focus on a specific pillar. For example, the items 3.2 – Customer integration, 5.7 - Digital platforms for customer integration and 7.4 - Internet of services contribute to the human-centric design pillar offering the company the possibility to better understand customers' needs and conceive customized products and services (Or et al., 2023; Shifrin and Michel, 2022). The items 4.2 – Real-time production monitoring and control, 5.2 - Software systems like ERP, MES, CRM and PLM, 6.3 - Blockchain Technology, 7.1 - Cloud Computing (CC) network for resources sharing and 7.2 - Cloud Computing (CC) network for data storing contribute to the resilience pillar by providing the company with the ability to improve decision-making based on real data (Nassif et al., 2010; Semeraro et al., 2023), consult integrated information in real-time and remotely (Gupta et al., 2023), and support internal and external traceability (Leng et al., 2022), resulting in a more resilient production system (Van Oudenhoven et al., 2023). The item 6.1 - Additive Manufacturing (AM), 3D Printing (3DP) is useful for helping the company reduce material waste (Ivanov, 2023) energy consumption in production (Oláh et al., 2020), or implement sustainable packaging. In this context, and based on the reasons discussed in Section 3.1, two types of measurement statement were proposed: the first for the integrated assessment of I5.0 (IA), and the second for the I5.0 pillar-based assessment (PBA).

4.4. Roadmap for improvement

After the assessment phase of the company's maturity level in I5.0 and the related pillars, the proposed MM is useful for providing guidelines to support the company improvement. Fig. 5 presents 23 guideline archetypes that represent a roadmap for companies aiming to improve their maturity level across the seven dimensions of the proposed I5.0 MM. These can be considered as ideas for formulating specific actions for the relevant business case, helping the manager or the business analyst navigate the improvement path toward the three pillars of I5.0. For each dimension, this roadmap proposes an archetype of card in which the company can collect useful information about:

- The timeline for implementation, which includes information about the stage of guideline implementation: the initial stage of analysis in which the company starts the improvement project and understand the requirements; the design stage, in which the company plans the improvement path; the execution stage, in which the company implements the improvement path; and the monitoring stage, in which the company assesses the new maturity level.
- The resources, which gather information about the type of resources needed to implement the guidelines (e.g., financial, internal or external personnel, technological, intellectual).
- The KPIs, which provide information about indicators useful for quantitatively assessing the progress of the improvement path, stage by stage. These can also serve as a point of comparison with competitors involved in similar improvement paths or simply as means of better tracking the process.
- The I5.0 pillars, which include information about the various pillars involved in the improvement path. As explained in Table 2, each dimension can be implemented from three different viewpoints of

Table 3
- The proposed MM for I5.0.

Dimension *(FAHP dimension weights)*	Measurement Item *(FAHP item weights)*	Measurement statement *(IA – Integrated Assessment; PBA – I5.0 pillar-based assessment)*	I5.0 Pillars *S: sustainability,* *R: resilience,* *H: human-centered design*
1 – People and culture (0,39)	1.1 – Leadership support (0,46)	***IA*** - The leadership of my company promote and support digital transformation journey for a sustainable, resilient and human centered production. ***PBA*** – The "leadership support" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	1.2 – Continuous improvement culture (0,27)	***IA*** - Following the continuous improvement culture viewpoint, my company promote the adoption of technology-enabled sustainable, resilient and human centered production. ***PBA*** – The "continuous improvement culture" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	1.3 – Dedicated teams (0,18)	***IA*** - In my company there are work teams focused on the implementation of technology-enabled sustainable, resilient and human centered production. ***PBA*** – The "dedicated teams" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	1.4 – Digital skills (0,09)	***IA*** - The employees of my company have suitable digital skills to implement a technology-enabled sustainable, resilient and human centered production. ***PBA*** – The "digital skills" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
2 – Awareness on I5.0 production (0,19)	2.1 – Familiarity (0,39)	***IA*** - I am familiar with the concept of technology-enabled sustainable, resilient and human centered production. ***PBA*** – The "familiarity" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	2.2 – Impact perception (0,32)	***IA*** - I have a perception of the impact generated by technology-enabled sustainable, resilient and human centered production. ***PBA*** – The "impact perception" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H

(*continued on next page*)

Table 3 (*continued*)

Dimension *(FAHP dimension weights)*	Measurement Item *(FAHP item weights)*	Measurement statement *(IA – Integrated Assessment; PBA – I5.0 pillar-based assessment)*	I5.0 Pillars *S: sustainability,* *R: resilience,* *H: human-centered design*
	2.3 – Usefulness (0,16)	***IA*** - I perceive the usefulness of technology-enabled sustainable, resilient and human centered production in improving company performance. ***PBA*** – The "usefulness" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	2.4 - Readiness for adoption (0,13)	***IA*** - I think I am ready to introduce in my company technologies to make the production sustainable, resilient and human centered. ***PBA*** – The "readiness for adoption" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
3 - Organizational strategy (0,16)	3.1 – Digital vision and roadmap (0,44)	***IA*** - My company adopt a digital vision and a roadmap to guide the implementation of technology-enabled sustainable, resilient and human centered production. ***PBA*** – The "digital vision and roadmap" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	3.2 – Customer integration (0,23)	***IA and PBA*** -My company, in the process of product or service development, take into account the consumers' needs	H
	3.3 – Collaboration (0,14)	***IA*** - My company has collaboration with external organization (e.g., technology providers, consultants, universities) to conceive and implement technology-enabled sustainable, resilient and human centered production. ***PBA*** – The "collaboration" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	3.4 – Zero paper strategy (0,09)	***IA*** - My company do not use print paper during control, visualization and data exchange processes. ***PBA*** – The "zero paper strategy" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%.	S R
	3.5 – Financial investment (0,08)	***IA*** - My company invests in technology-enabled sustainable, resilient and human-centered production. ***PBA*** – The "financial investment" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H

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Table 3 (*continued*)

Dimension *(FAHP dimension weights)*	Measurement Item *(FAHP item weights)*	Measurement statement *(IA – Integrated Assessment; PBA – I5.0 pillar-based assessment)*	I5.0 Pillars *S: sustainability,* *R: resilience,* *H: human-centered design*
4 - Value chain and processes **(0,10)**	4.1 – Digitalization of vertical value chain (0,43)	***IA*** - In my company, vertical integration in value chain is supported by technologies that allows to achieve resilient and human centered production aims (vertical integration is when the company controls the upstream or downstream stage along its supply chain). ***PBA*** – The "digitalization of vertical value chain" maturity score of your company is "_": ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	R H
	4.2 – Real-time production monitoring and control (0,25)	***IA and PBA*** -My company, production processes are monitored in real-time reacting dynamically to changes in order to assure resilient production.	R
	4.3 - End-to-end IT-enabled planning and steering process (0,12)	***IA*** - My company uses technologies for sustainable and resilient end-to-end planning and steering processes (such as selling forecasting, production planning, warehouse and logistics planning). ***PBA*** – The "end-to-end IT-enabled planning and steering process" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%.	S R
	4.4 – Digital production equipment (0,13)	***IA*** - My company uses digital equipment to improve the levels of sustainability, resilience and human centricity of the production (such as sensors; IoT; digital monitoring, control, optimization and automation). ***PBA*** – The "digital production equipment" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	4.5 – Digitalization of horizontal value chain (0,07)	***IA*** - In my company, horizontal integration is supported by technologies that allow to achieve in production the aims of resilience and human centricity (horizontal integration is when the company control other companies in the same stage of the supply chain). ***PBA*** – The "digitalization of horizontal value chain" maturity score of your company is "_": ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	R H
5 - Smart manufacturing technology **(0,07)**	5.1 – Autonomous and collaborative robots (0,21)	***IA*** - My company uses advanced robots for automating production activities (such as handling, washing, packaging) and/or autonomous vehicles for logistics activities. These systems work autonomously, and interact collaboratively with human work and the surrounding environment with the aim of making the production more resilient and human centered. ***PBA*** – The "autonomous and collaborative robots" maturity score of your company is "_": ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	R H
	5.2 - Software systems like Enterprise Resource Planning (ERP), Manufacturing Execution System (MES), Customer Relationship Management (CRM) and Product Lifecycle Management (PLM) (0,25)	***IA and PBA*** - My company uses digital software to share information between the several company areas and obtain feedback useful to take decisions for resilient production.	R

(*continued on next page*)

Table 3 (*continued*)

Dimension *(FAHP dimension weights)*	Measurement Item *(FAHP item weights)*	Measurement statement *(IA – Integrated Assessment; PBA – I5.0 pillar-based assessment)*	I5.0 Pillars *S: sustainability,* *R: resilience,* *H: human-centered design*
	5.3 – Identifiers like bar code, QR code (Quick Response code), RFID (radio frequency identification) and RTLS (Real-Time Locating System) (0,16)	***IA*** - My company uses identifiers for production assets (e.g., products, parts, tools or people) tracking and locating it in the various areas of the company such as manufacturing plants, warehouse and maintenance. ***PBA*** – The "Identifiers like bar code, QR code, RFID and RTLS" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%.	S R
	5.4 – Sensors, actuators, embedded systems and PLCs (Programmable Logic Controller) (0,08)	***IA*** - My company adopts sensors, actuators and PLC on machines and equipment to make the production more sustainable and resilient. ***PBA*** – The "sensors, actuators, embedded systems and PLCs" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%.	S R
	5.5 - Machine to Machine (M2M) and Human to Machine (H2M) communication (0,07)	***IA*** - My company has interoperable communication systems to exchange information among production machines and employees in various hierarchical levels with the aim to improve the production resilience and human centricity. ***PBA*** – The "M2M and H2M" maturity score of your company is "_": ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	R H
	5.6 – Digital platforms for supplier integration (0,07)	***IA*** - My company uses digital platforms to exchange real-time information on production performance among production units, suppliers and warehouse to improve the production sustainability, resilience and human centricity. ***PBA*** – The "digital platforms for supplier integration" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	5.7 – Digital platforms for customer integration (0,07)	***IA and PBA*** - My company uses digital platforms to share with consumers information about production performance and provide support to specific requests.	H
	5.8 – Augmented Reality (AR), Virtual Reality (VR) and Mixed Reality (MR) (0,06)	***IA*** - My company uses technological devices of augmented reality to improve the sustainability, resilience and human centricity of production (such as virtual visualization of problems in product development phases, virtual glasses to support employees in decisions). ***PBA*** – The "AR, VR, and MR" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
6 – Technology based products and services **(0,06)**	6.1 - Additive manufacturing (AM), 3D Printing (3DP) (0,39)	***IA and PBA*** - My company uses 3DP techniques to improve sustainability in production (such as reducing waste in materials, realizing sustainable packaging, reducing production energy).	S

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Table 3 (*continued*)

Dimension *(FAHP dimension weights)*	Measurement Item *(FAHP item weights)*	Measurement statement *(IA – Integrated Assessment; PBA – I5.0 pillar-based assessment)*	I5.0 Pillars *S: sustainability,* *R: resilience,* *H: human-centered design*
	6.2 – Mobile and wearables devices (0,32)	***IA*** - My company uses mobile (e.g., smartphones e tablet) wearables devices (e.g., smart watch, smart glass and smart gloves) to consult information about the production and to communicate in real-time with different systems. ***PBA*** – The "mobile and wearables devices" maturity score of your company is "_": ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	R H
	6.3 - Blockchain Technology (BT) (0,12)	***IA and PBA*** - My company uses BT to make resilient the supply chain (e.g., providing traceability information) and production (e.g., certifying sustainability indicators).	R
	6.4 - Smart Product (0,19)	***IA*** - My company produces smart products with sensors capable to interact the surrounding environment with the aim to improve the production sustainability, resilience and human centricity. ***PBA*** – The "smart product" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
7 – Industry 4.0 Technologies (0,05)	7.1 - Cloud Computing (CC) network for resources sharing (0,27)	***IA and PBA*** - My company uses CC systems for remote connection and sharing of hardware (e.g., equipment, robots) and software (e.g., data, documents, software) resources improving the production resilience.	R
	7.2 - Cloud Computing (CC) network for data storing (0,16)	***IA and PBA*** - My company storages and retrieves information from CC network with the aim of improving production resilience.	R
	7.3 - Internet of Things (IoT) (0,13)	***IA*** - My company uses technologies for wireless communication among machines, robots, IT systems and employees (e.g., Wi-Fi, 0.17 ZigBee, and Bluetooth) that are capable of improving the sustainability, resilience and human centricity of production. ***PBA*** – The "Internet of Things" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	7.4 - Internet of Services (IoS) (0,11)	***IA and PBA*** - My company provides, to customers and suppliers, services based on web technologies capable of adding customized value.	H
	7.5 - Processing Big Data (BD) in real-time (0,13)	***IA*** - My company is capable of collecting, storing and managing data bases (with data from physical objects and external data sources) in real-time with the aim of improving the sustainability, resilience and human centricity of production plant. ***PBA*** – The "processing BD in real-time" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	7.6 – Simulation tools (0,09)	***IA*** - My company uses production data to simulate possible scenarios evaluating several solutions (or problems) in production processes. ***PBA*** – The "simulation tools" maturity score of your company is "_": ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H

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Table 3 (*continued*)

Dimension (*FAHP dimension weights*)	Measurement Item (*FAHP item weights*)	Measurement statement (*IA – Integrated Assessment; PBA – I5.0 pillar-based assessment*)	I5.0 Pillars *S: sustainability,* *R: resilience,* *H: human-centered design*
	7.7 - Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL) (0,06)	***IA*** - My company uses technologies based on artificial intelligence (e.g., natural language processing, speech recognition, rule-based systems and computer vision) useful to improve products, services and production. ***PBA*** – The "AI, ML and DL" maturity score of your company is " _ "; ▪ This value can be attributed to the achievement of sustainability objectives for _%. ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	S R H
	7.8 – Industrial Cyber Security (CS) (0,05)	***IA*** - My company uses several IT security measures (e.g., encryption, authentication and authorization) that improve communication security and data protection. ***PBA*** – The "industrial CS" maturity score of your company is " _ "; ▪ This value can be attributed to the achievement of resilience objectives for _%. ▪ This value can be attributed to the achievement of human-centered design objectives for _%.	R H

production: sustainability, resilience and human-centered. However, the company can choose to address only one or more pillars in the specific improvement actions.

The archetype of cards can be updated over time according to the different development stages, incorporating new resources, I5.0 pillars or KPIs as the improvement path matures. Details about how to use the proposed Roadmap, along with the related guidelines and card archetypes, are provided in the description of the multiple case studies (Section 5).

5. Empirical evidences from a multiple case study

5.1. Case study 1: 3D pasta producer

The data collection phase in the context of case study 1 (3D pasta producer) involved administering the questionnaire to three employees of the company with the following profiles: marketing manager, sales specialist, production manager. After analyzing the first set of data, in-depth interviews were conducted to further investigate the company's performance along the three I5.0 pillars, involving the same employees. The company is a limited liability company (LLC), with fewer than 10 employees and a turnover of <2 million. The respondents stated that the company is highly technological (2 out of 3 respondents) and fairly sustainable, resilient and human-centered (3 out of 3 respondents) compared to other companies in the same sector. They also indicated that the company uses technology sufficiently to improve the sustainability, resilience and human-centered of production. Fig. 6 shows the maturity score found for each maturity dimension (in light color) and the overall maturity score (in dark color) for the integrated (in black color) and I5.0 pillar-based assessments (in green, yellow and blue colors).

It emerges that the least developed maturity dimension is related to "Technology-based products and services" (6), with a dimension score of 2,91. In contrast, the most developed dimension is related to "Awareness of I5.0 production" (2), in which the company achieves a maturity score of 4,35, followed by the dimension "People and culture" (1) with a maturity score of 4,02. The remaining dimensions - "Organizational strategy" (3) with 3,41; "Value chain and processes" (4) with 3,78; "Smart manufacturing technology" (5) with 3,06; and "Industry 4.0 technologies" with 3,30 - are positioned on an average score. This allowed the company to obtain an overall maturity level of 3,87, which places it at Level 3: advanced. Therefore, I can conclude that the company has a good level of maturity in implementing an I5.0 production model based on the use of technology. However, upon deeper analysis, it is interesting to note how the three pillars of I5.0 are not equally mature. Although no single dimension clearly outweighs the others, it is worth noting that human-centered design is the least developed. With reference to the least developed maturity dimension, "Technology-based products and services" (6), it is primarily developed to achieve sustainability objectives (Sustainability maturity score = 1,92), followed by resilience (Resilience maturity score = 0,7) and human-centered design (human-centered design maturity score = 0,32) objectives. A similar trend is observed in the most developed dimension "Awareness of I5.0 production" (2), which is developed first to achieve sustainability objectives (Sustainability maturity score = 2,07), followed by resilience (Resilience maturity score = 1,2) and human-centered design objectives (Human-centered design maturity score = 1,16). However, in this case, the gap between resilience and human-centered design is smaller. It is also interesting to note that only the dimension "Organizational strategy" (3) is more advanced in the human-centered design pillar compared to the other two pillars.

Going deeper into the survey, Fig. 7 illustrates the maturity score recorded for each item in each dimension for both the Integrated (in grey color) and I5.0 pillar-based assessments (in green, yellow and blue color).



Fig. 4. - Case studies selection, motivations.

Although dimension (6) "Technology based products and services" has the lowest score (2,91), it includes one of the items with the highest maturity score (6.1 - Additive manufacturing (AM), 3D printing (3DP) – 4,67). This item is designed to primarily address the sustainability pillar. The product offered by the company is made through a production process based on 3D printing, which, however, is the only technological element included in the product concept. Interestingly, items "2.3 - Usefulness" and "3.2 - Customer integration" also register high maturity score (4,67), respectively reflecting the company's strong awareness of the perceived usefulness of sustainable, resilient and human-centered production based on the use of technologies, as well as its ability to consider the consumer's needs during the product development process. Indeed, the company develops specific pasta formats to meet the needs of starred chefs and restaurants and launches themed pasta formats designed to cater to end consumers at different times of the year (e.g., Christmas, Easter, Valentine's Day, Mother's Day). Analyzing the "2.3 - Usefulness" item across the three I5.0 pillars, it is primarily developed toward sustainability objectives (2,33) and equally developed toward resilience and human-centered design objectives (1,17). Instead, the "3.2 - Customer integration" item is designed to primarily address the human-centered design pillar.

On the other hand, items with the lowest maturity values turn out to be: "4.5 – Digital horizontal integration of value chain - 2"; "5.8 – AR, VR and MR – 1,67"; "6.2 – Mobile and wearable devices– 1,67"; "6.3 - Blockchain Technology (BT) – 2"; "6.4 - Smart Product – 1,67".

Following this analysis and referring to the roadmap for improvement (Fig. 5), a set of guidelines was developed to help the company improve its maturity level in I5.0, with the goal of enhancing the resilience and human-centric pillars. These guidelines were conceived and discussed with the marketing and production managers of the company to create a concrete path for implementing I5.0. It emerged that the company is improving its offerings through a process of servitization. Servitization is defined as the transformational process of shifting from a product-centric business model to a service-centric approach (Kowalkowski et al., 2017), which can lead the company to leverage technologies to enrich the product with information that can be conveyed to the end consumer (Xing et al., 2022). The company uses identifiers and sensors in its production processes that help make production more sustainable, resilient and human-centered. However, this information remains confined within the company's boundaries and is not widely communicated to the end consumer. Using this available information to create new product storytelling represents an opportunity to foster customer involvement in the company's activities, which can benefit customer satisfaction and trust. This, in turn, would improve

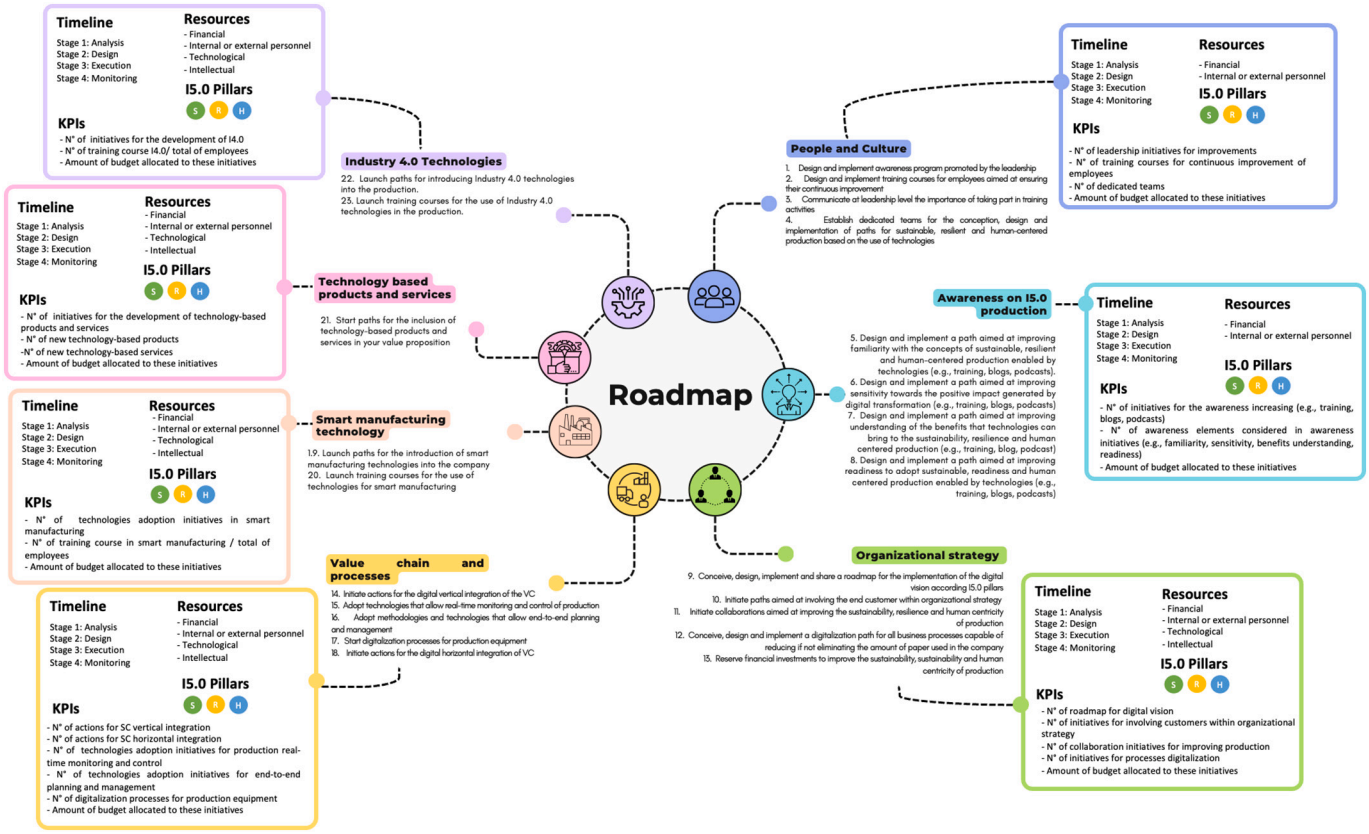


Fig. 5. Roadmap for improvements toward the I5.0 pillars.

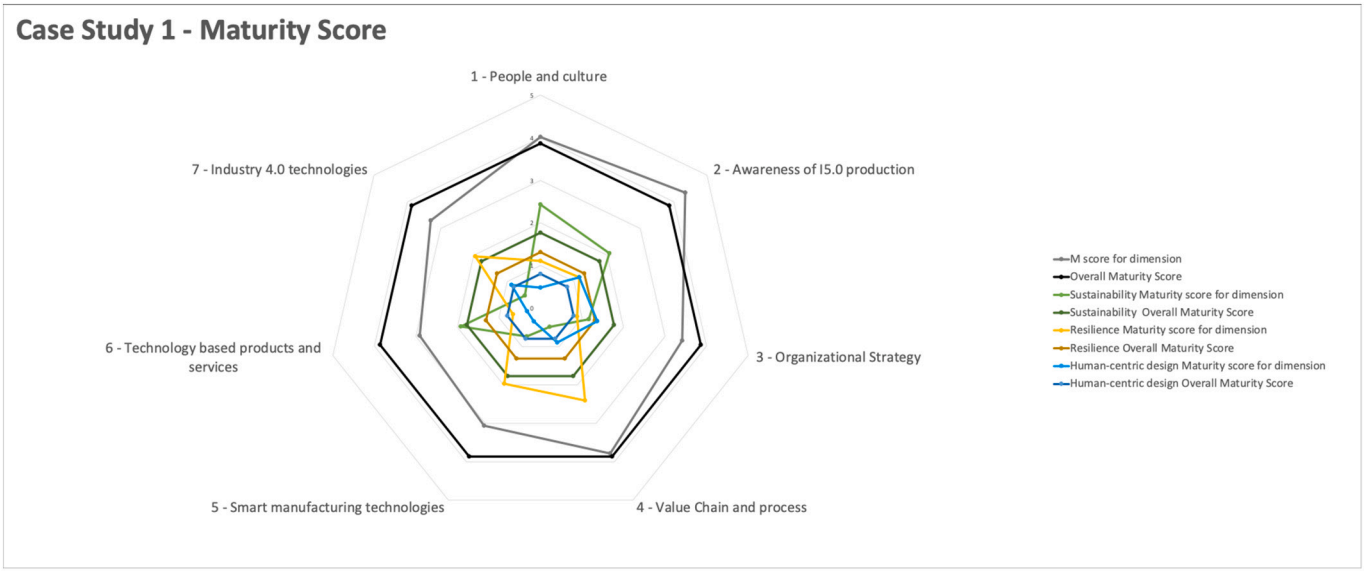


Fig. 6. - Maturity model, case study 1.

the human-centered pillar, which was identified as the last mature. By addressing dimension "6 - Technology-based products and services", the company is transforming its pasta packaging into smart packaging capable of transferring product and process information to the end consumer (6.4 - Smart Product). This information could be certified using blockchain mechanisms (6.5) to increase its authenticity and activate value- and risk-sharing mechanisms in B2B sales. The company is currently executing this action and has already allocated financial and technological resources to the project. Moreover, the internal personnel (market team) are working with external personnel to better understand how to communicate the collected information effectively. In addition, the adoption of technological devices based on virtual and augmented reality (5.8) could lead to virtual prototyping paths, useful for evaluating new pasta formats by involving the end consumer in immersive experiences, thereby increasing the human-centered design of production. The company is currently in the design phase of this action and has already allocated financial and technological resources to this project. Internal personnel are involved in conceiving new pasta formats to



Fig. 7. - Items maturity score, case study 1.

commercialize and the marketing team is involved in the selection of experimentation target groups. These efforts are supported by external personnel involved in the development of technological solutions based on virtual reality. However, the analysis revealed that the company has not yet implemented training course in virtual and augmented reality, which are essential to fully realize the potential of this project. Therefore, it is necessary to enhance the improvement strategy. Finally, the on-farm adoption of technologies for horizontal integration (4.5) could contribute to achieving resilient and human-centered goals in production. For instance, this integration could involve traditional pasta producer connected to the company, aiming to share processes, production steps, or know-how that are different yet complementary within the supply chain in which the company operates. This represents a new and promising improvement opportunity for the company, which has decided to follow the guideline by starting with a requirements analysis and evaluating the effort needed in terms of financial, personnel and technological resources. As of now, the budget for this project has not yet been allocated. Fig. 8 summarizes the roadmap for improvement in Case Study 1.

5.2. Case study 2: Vegetable and fruit producer

The data collection phase for Case Study 2 (vegetable and fruit producer) consisted of administering the questionnaire to three employees of the company with the following profiles: president and legal representative, employees (2). After analyzing this initial set of data, in-depth interviews were conducted with the same employees to explore the three I5.0 pillars maturity. The company is a cooperative, with 10 to 50 employees and turnover ranging between 10 and 50 million euros. The respondents indicated that the company is moderately advanced in terms of technology adoption and is average in sustainability, resilience and human-centered design (as stated by all three respondents) compared to other companies in the industry. They also reported that the company makes moderate use of technology to improve the sustainability, resilience and human-centered design of its production processes. Fig. 9 illustrates the maturity score obtained for each dimension (in light color) and the overall maturity score (in dark color) for the I5.0 integrated (in grey color) and I5.0 pillar-based assessments (in green, yellow and blue colors).

Also in this case, it emerges that the least developed dimension is the one related to “6 – Technology based products and services” with a maturity score of 1,66. Below the average score is also the dimension related to “7 - Industry 4.0 technologies” with a maturity score of 2,98. Again, the most developed dimension is the one related to “2 - Awareness of I5.0 production”, in which the company achieves a maturity score of 4,18. The remaining dimensions, “1 - People and Culture” with 3,48; “3 - Organizational strategy” with 3.61; “4 - Value chain and processes” with 3,29; and “5 - Smart manufacturing technologies” (5) with 3,23, are placed on an average score. This gave the company an overall maturity level of 3,48, which places it at Level 3: advanced. Therefore, we can conclude that the company has a good level of maturity in implementing a production model based on the use of I5.0.

Focusing on the three pillars, the company demonstrates a reasonably balanced level of maturity across sustainability (1,15), resilience (1,24) and human-centered design (1,4). With reference to the least developed maturity dimension "Technology-based products and services" (6), this dimension is primarily developed to achieve sustainability (Sustainability maturity score = 0,74) and resilience (Resilience maturity score = 0,64) objectives, but it is less mature in achieving human-centered design (human-centered design maturity score = 0,32) objectives. In contrast, the maturity dimension "7 - Industry 4.0 technologies" is predominantly developed along resilience objectives (1,74) and is less advanced in human-centered design (0,74) and sustainability (0,50). Referring to the most developed dimension "Awareness of I5.0 production" (2), this dimension is primarily developed to achieve sustainability objectives (Sustainability maturity score = 1,71) and equally developed for resilience (Resilience maturity score = 1,24) and human-centered design (Human-centered design maturity score = 1,24).

Going deeper into the survey conducted, Fig. 10 illustrates the maturity score recorded for each item in each dimension for the I5.0 integrated (in grey color) and I5.0 pillar-based assessments (in green, yellow and blue colors).

As described above, the dimension "6 - Technology based products and services" is the dimension with the lowest score (2,91), and includes three items with the lowest maturity score: "6.1 - Additive manufacturing (AM), 3D printing (3DP)" (1,33); "6.2 - Mobile and wearable devices" (1,33); and "6.4 - Smart product" (1,67). Nevertheless, it is noteworthy that the item "6.3 - Blockchain technology" has maturity score of 3.33. This highlights the company's pilot project involving the use of blockchain and a QR code on the product labels to communicate and certify transparency and traceability data 2 for two product lines (tomato and watermelon). This item is designed to measure the resilience pillar. Low maturity values were also recorded for items "3.4 – Zero paper strategy – 2"; "5.1 - Autonomous and collaborative robots - 2,33"; "7.7 - Artificial intelligence (AI), machine learning (ML) and deep learning (DL) - 1,33"; and "7.8 - Industrial Cyber Security (CS) - 2,33".

On the other hand, with the highest maturity scores, there was the following items: "2.2 – Impact perception - 4.33"; "2.3 – Usefulness - 4,67", "3.2 - Customer integration - 4,33", "3.3 – Collaborations - 4,67" attesting to a high degree of company awareness toward the utility and perception of the impact that I5.0 production generates on business performance, and the importance of the customer and external collaborations in the production stages. Specifically, the items belonging to the second dimension (2.2 and 2.3) were found to be equally mature across the three pillars. However, the item "3.3 – collaboration" demonstrated greater maturity along the human-centered design pillar (3,27).

Following this analysis, and leveraging the guideline archetypes outlined in the roadmap for improvement (Fig. 5), a set of guidelines was developed to enhance the company's maturity in I5.0 production, with a focus on improving the three I5.0 pillar. These guidelines were conceived and discussed with the president and legal representative of the company to establish a concrete path for implementing I5.0.

The company is enhancing its production capacity by introducing autonomous systems for product washing and packaging, aimed at minimizing production waste ("5.1 - Autonomous and Collaborative Robots") and improving the resilience and human-centred design of production processes. Current greenhouse efficiency processes, based on sensor technology (5.4 - Sensors, actuators, embedded systems and PLCs) and identification technology (5.3 - Identifiers like bar code, QR code, RFID and RTLS), could be further integrated with the pilot transparency and traceability system. This system narrates product and production details to the end customer, with the potential to extend the use of sensor technology and blockchain to all production lines. Additionally, the strengths of the company's human-centered production, as demonstrated by high maturity score in "3.2 - Customer Integration - 4,33" and "3.3 – Collaborations - 4,67", could be further leveraged through the adoption of digital platforms for supplier and customer integration (5.6 and 5.7). This effort would significantly enhance the current average maturity value of the dimension "5 - Smart Manufacturing Technologies". This wide-ranging project is currently in different stage of implementation. For instance, autonomous systems are in the design phase, while the integration of sensors and identification technologies in the pilot transparency and traceability system is in the execution phase. The company has already allocated financial, technological and personnel resources (both internal and external) for these initiatives. Furthermore, the company could initiate a project to reduce the use of printed paper ("3.4 – Zero paper strategy") in both internal

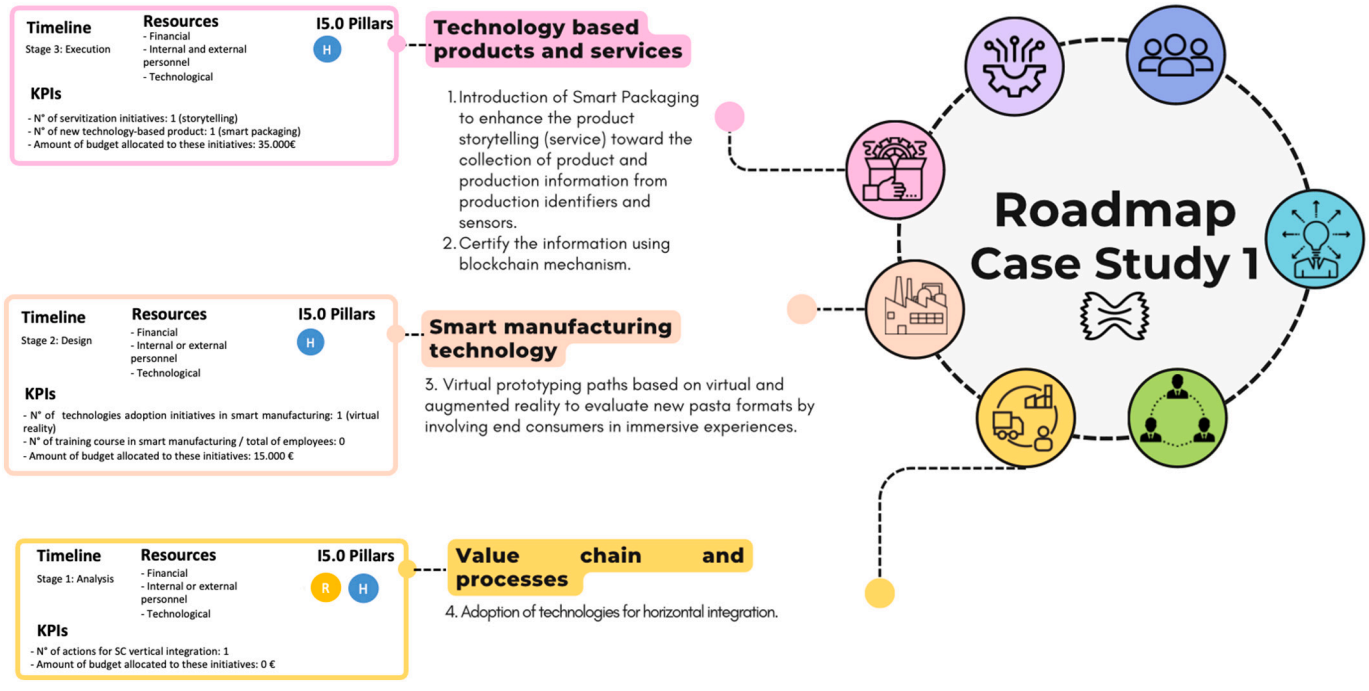


Fig. 8. - Improvement roadmap, case study 1.

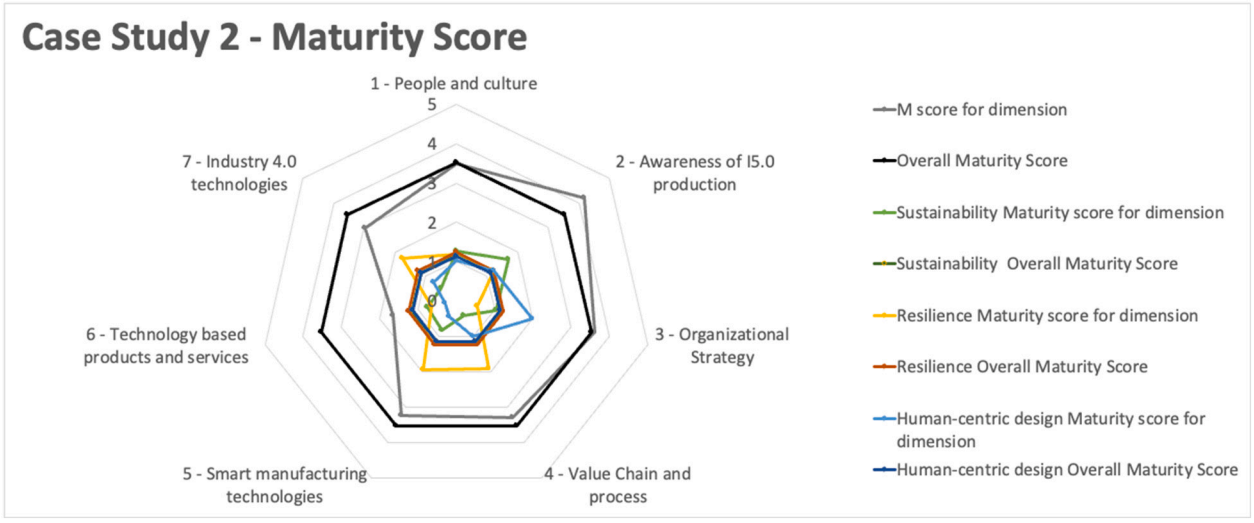


Fig. 9. – Maturity model, case study 2 (Note: the sustainability overall maturity score is overlapped with the human-centric design overall maturity score and therefore not visualizable in this graph).



Fig. 10. - Items maturity score, case study 2.

operational processes and external activities, promoting products through smart and virtual labels. Improving this maturity item would bring sustainable and resilience benefits. This represents a new I5.0 project, currently in the first stage of the timeline (analysis). During this stage, the company is identifying internal personnel, as well as technological and financial resources, necessary for project implementation. Fig. 11 summarizes the roadmap for improvement in the Case Study 2.

5.3. Case study 3: Shoe producer

The data collection phase for Case Study 3 (shoe producer) involved administering a questionnaire to three employees with the following profiles: owner and employee (2). Following the analysis of this first set of data, in-depth interviews were conducted with the same employees to investigate the company's practices along the three I5.0 pillars. The company is a Limited Liability Company (LLC), with 50 to 250

employees and a turnover between 2 and 10 million euros. Respondents stated that the company is relatively advanced in terms of technology, sustainability, resilience and human centered design (2 out of 2 respondents), when compared to other companies in the same sector. However, they also indicated that the company makes limited use of technology to enhance the sustainability, resilience and human-centered design of production processes. Fig. 12 illustrates the maturity score for each dimension (in light color) and the overall maturity score (in dark color) for the I5.0 integrated (in grey color) and I5.0 pillar-based assessments (in green, yellow and blue colors).

The company records a low overall maturity score in I5.0-based production (2,26) corresponding to Level 2: beginner. The lowest-ranking dimension is "7 - Industry 4.0 technologies" with a score of 1,46, followed by dimensions "6 - Technology based products and services" with 1,85″ and "3 - Organizational strategy" with 1.91. Even the dimensions with higher scores show only average maturity: "1- People and culture" achieves a score of 2,54, while "4 - Value chain and processes" with 2,46. From these results, I can conclude that the company is in the initial stages of implementing a production model based on I5.0 principles. Going deeper into the analysis, the resilience pillar of I5.0 emerged as the most mature (0,86), followed by human-centered design (0,76) and sustainability (0,59). This trend is also evident in the lower-scoring dimension, "7 - Industry 4.0 technologies" and "6 - Technology based products and services". Focusing on the more mature dimensions, "1- People and culture" was found to be more developed along sustainability (0,93) and resilience (0,95) pillars. Similarly, "4 - Value chain and processes" is mainly developed along the resilience pillar (1,66) but show a very little development along the sustainability pillar (0,26).

Delving further into the survey results, Fig. 13 illustrates the maturity scores recorded for each item in each dimension for the I5.0 integrated (in grey color) and I5.0 pillar-based assessments (in green, yellow and blue colors).

In the case of the integrated assessment, the item maturity scores diagram assumes a peculiar pattern, recording minimum values for a large number of items in all dimensions and, in a mirror image, presenting maximum values for some of the other items in the same dimensions. This behavior can be observed also in the I5.0 pillar-based assessment, in which high value were registered only for item focused to assess a single pillar (e.g., 3.2).

This trend confirms the earlier observation regarding the company's initial steps on the path of digital production. It is noteworthy that company leadership (1.1) appears to be moderately involved in the digital transformation aimed at sustainable, resilient and human-centered design production, and there is evidence of a continuous improvement culture (1.2). However, within the statements for dimension "1 - People and Culture", the company has not yet established dedicated teams for the implementation of I5.0-based production (1.4) and believes that employees' digital skills are low (1.4). This is reflected by a score close to the minimum value, although these skills contribute to achieving maturity across the three pillars in a fairly balanced manner.

An intriguing finding pertains to dimension "3 - Organizational strategy", where all maturity items scored near the minimum value, except for "3.2 - Customer integration", which recorded a score close to the maximum value (4,8). This item is recognized as addressing only the human-centered design pillar. On the technological side, very low maturity scores were observed for both "5- Smart manufacturing technologies" and "7 - Industry 4.0 technologies", which have been implemented primarily with a focus on the human-centered design pillar. Notable exceptions include "5.5 - M2M and H2M technologies" (3,8) and "7.3 - IoT" (3,9), which appear to be used on moderately within the enterprise, without exhibiting dominance toward a single I5.0 pillar. Similar situation exists within dimension "6 – Technology based products and services", where the company reports using mobile devices for production monitoring but shows low levels of adoption for AM, 3DP, blockchain and smart product technologies. The primary focus of these implementations is directed toward achieving of human-centered design objectives.

Based on this analysis and leveraging the guideline archetypes reported in the roadmap for improvements (Fig. 5), a set of guidelines was developed to help the company enhance its maturity in I5.0 production, with the goal of advancing the three pillars of I5.0. These guidelines were conceived and discussed with the company's owner to establish a clear and actionable path for I5.0 implementation.

The roadmap for this company first includes guidelines aimed strengthening dimensions 1 and 2. It is considered strategic for any company engaged in digital transformation process to address the soft elements of the organization. Specifically, the establishment of dedicated teams for the implementation of I5.0 production - focused on leveraging technology for sustainable, resilient and human-centered purposes - was suggested. This initiative aims to promote the adoption of I5.0 principles and raise awareness among all employees about its overall benefits, as well as its specific advantages in terms of sustainability, resilience and human-centered design. Additionally, it is crucial to initiate employees training programs to enhance the digital skills and qualifications of staff (1.4), preparing them for the increased use of technology in future operations. To foster acceptance of change, it is essential to clearly communicate the goals of the digitalization process. This involves raising awareness about the positive impact that such production can have on workers' well-being, society, the environment

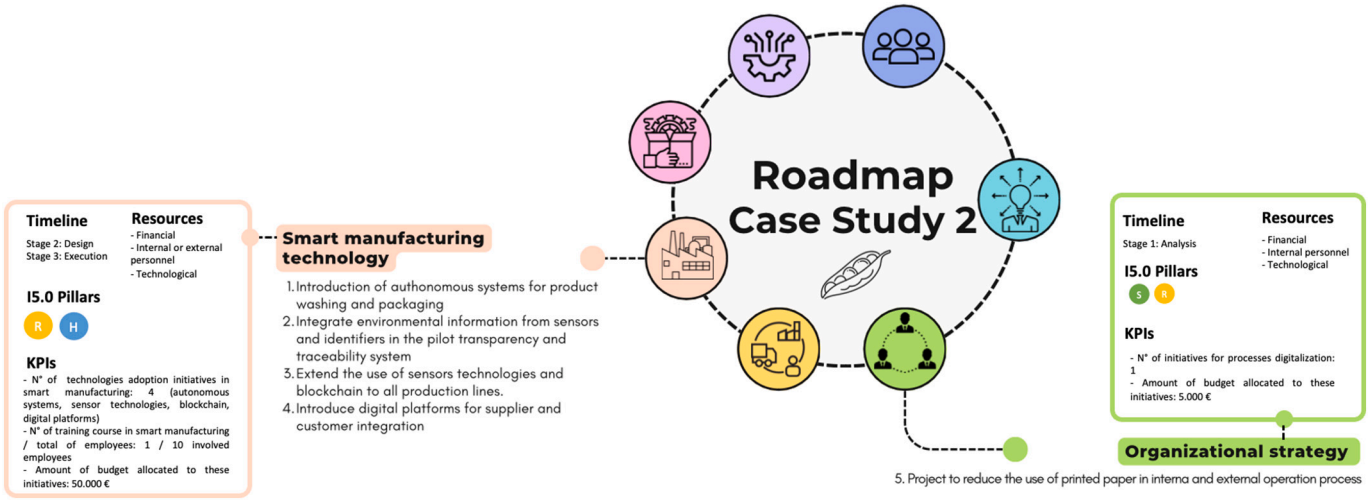


Fig. 11. - Improvement roadmap, case study 2.

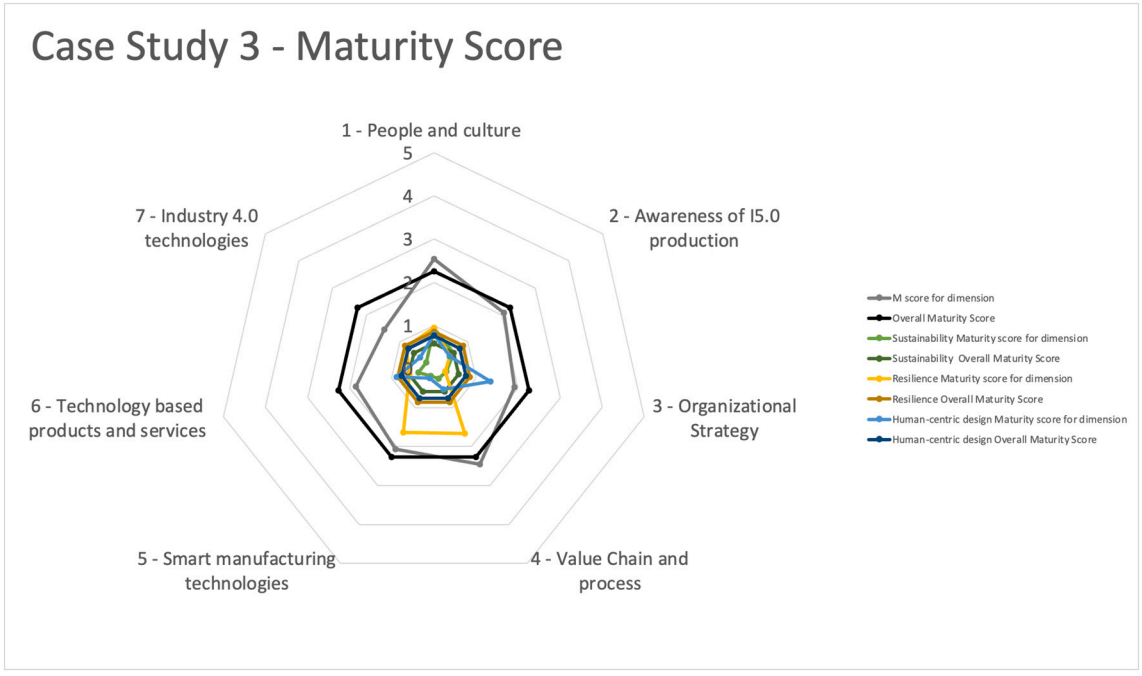


Fig. 12. – Maturity model, case study 3.

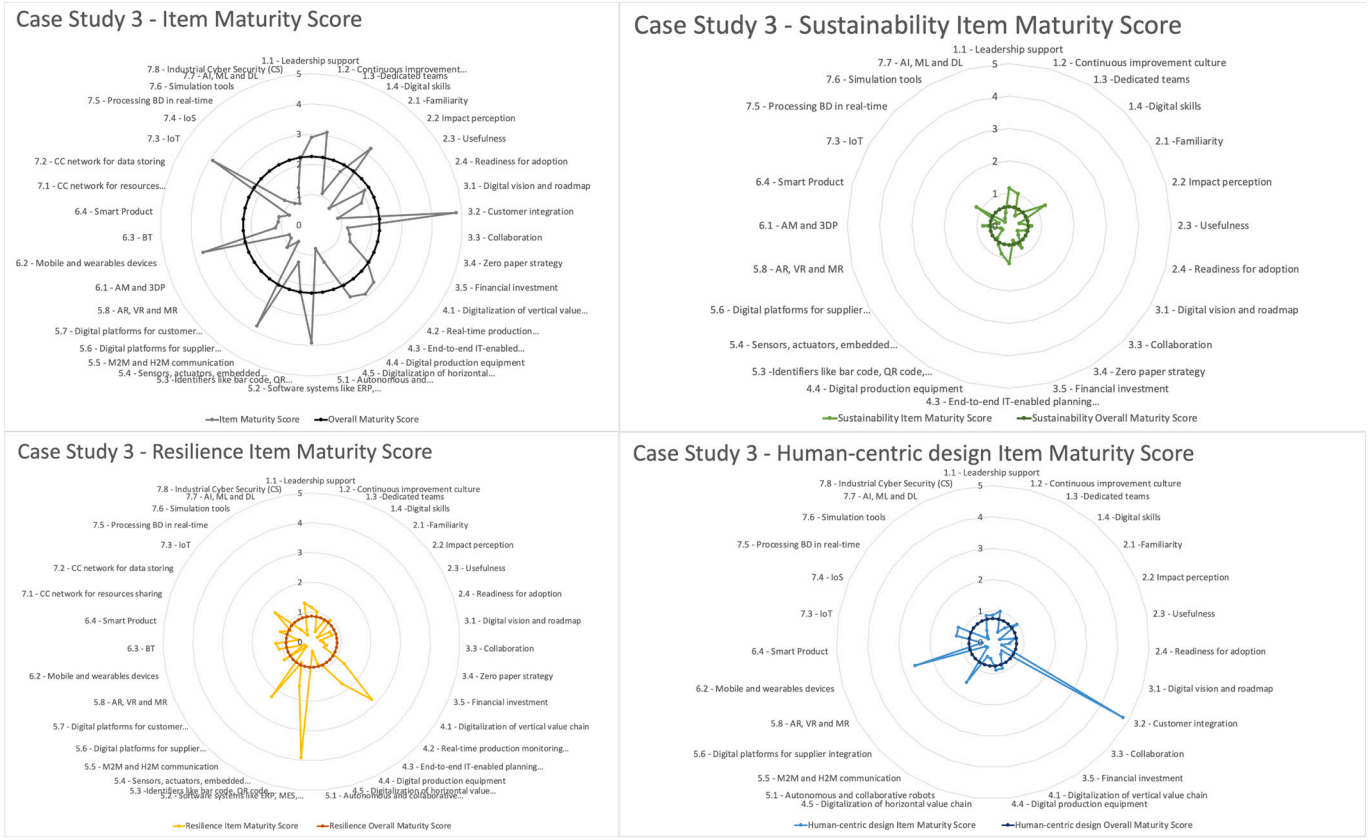


Fig. 13. - Items maturity score, case study 3.

and the company itself (2.3). These strategies are expected to contribute to improving the company's I5.0 maturity level, resulting in benefits for sustainable, resilient and human-centered production goals. However, these initiatives are still in the early stages of timeline. Although the company acknowledges the needs of personnel (both internal and external) and financial resources, no budget has yet been allocated.

Therefore, it is necessary to identify funding– either through internal or external sources - that can support the company in adopting new technologies. Fig. 14 summarizes the roadmap for improvement in the Case Study 3.

5.4. Case study 4: Bag producer

The data collection phase for Case Study 4 (bag producer) consisted of administering a questionnaire to three employees of the company with the following profiles: owner (2) and managing director. After analyzing this initial set of data, in-depth interviews were conducted with the same employees to further investigate the company's alignment with the e three I5.0 pillars. The company is a Limited Liability Company (LLC), with 11 to 50 employees and a turnover of <2 million euros. The respondents stated that the company is moderately technological, sustainable, resilient, and human centered (as indicated by 2 out of 3 respondents) compared to other companies in the same sector. However, they also noted that the company uses minimal technology to enhance sustainability, resilience and human-centered design aspects of production (2 out of 3 respondents). Fig. 15 illustrates the maturity scores recorded for each dimension (in light color) and the overall maturity score (in dark color) for the I5.0 integrated (in grey color) and I5.0 pillar-based assessments (in green, yellow and blue colors).

The company demonstrates a medium-high overall maturity level, registering a general maturity score of 3.65, which places it in at Maturity Level: advanced. The dimension with the lowest score is "6 – Technology based products and services - 2.17", reflecting the company's lack of technology-based product offerings. This is followed by "5 – Smart manufacturing technologies", which records a score of 2.99. All other dimensions achieve a maturity score between 3 and 4, with the highest score recorded in "1 -People and Culture - 3.75", indicating strong company awareness of issues related to sustainable, resilient and human-centered production based on I5.0 principles. Focusing on maturity across the three I5.0 pillars, the least developed dimensions, "6 – Technology based products and services" and "5 – Smart manufacturing technologies", are primarily aligned with resilience objectives, recording maturity score of 0,92 and 2,26, respectively. Conversely, the most developed dimension, "1 -People and Culture", demonstrates a more balanced distribution across the three pillars, with slightly lower emphasis on the resilience pillar (Sustainability dimension maturity score: 1,34; Resilience dimension maturity score: 1; Human-centered design dimension maturity score: 1,40).

Delving further into the survey results, Fig. 16 illustrates the maturity score for each item within each dimension for the I5.0 integrated (in grey color) and I5.0 pillar-based assessments (in green, yellow and blue colors).

As described above, the dimension "6 – Technology based products and services" has the lower overall the lowest score (2,17) and includes several items with the lowest maturity score: "6.1 - Additive manufacturing (AM), 3D printing (3DP) - 1,67"; "6.3 - Blockchain technology (BT) – 2"; and "6.4 - Smart Product - 1,67". Similarly, low maturity values were recorded for the following items "5.1 - Autonomous and collaborative Robots - 2,33", "5.4 - Sensors, actuators, embedded systems and PLCs - 2,00", "5.8 - AR, VR and MR - 2,00", and "7.7 - Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL) - 2,00". No evident trend was observed in the maturity levels of these low-maturity items across the different pillars.

On the other hand, the highest maturity scores were recorded for the following items: "1.1 - Leadership Support - 4,00"; "2.4 - Readiness for adoption - 4,00"; "3.2 - Customer Integration - 4,33", "5.2 - Software systems like ERP, MES, CRM and PLM - 4,00". Among these, "3.2 - Customer integration" is the highest-scoring item overall, highlighting the company's strong ability to involve customer in the product design process. From the perspective of the I5.0 pillars, same high-maturity items, such as "1.1 – Leadership support" and "2.4 – Readiness for adoption", exhibit equal maturity across the three pillars. In contrast, others address specific pillars, such as "3.2 – Customer integration" (focused on human-centered design) and "5.2 – Software systems" (aligned with resilience).

Following this analysis, and leveraging the guideline archetypes reported the roadmap for improvement (Fig. 5), a set of guidelines was developed to help the company enhance its maturity in I5.0 production. These guidelines were conceived and discussed with the company's owners to establish a concrete path for I5.0 implementation.

The improvement strategy includes a series of guidelines aimed at strengthening dimensions 5 and 6 while preserving the company's core vision of craftsmanship and product customization. Specifically, the integration of 3D printing technologies (6.1) into the production process is of interest to the company. This would allow for the design of accessories and special elements to enrich the product portfolio or improve existing products (e.g., customized 3D-printed keyrings, zipper sliders, etc.). Additionally, certain product lines could be equipped with smart sensors (6.4) to monitor conditions of use, such as sensors for

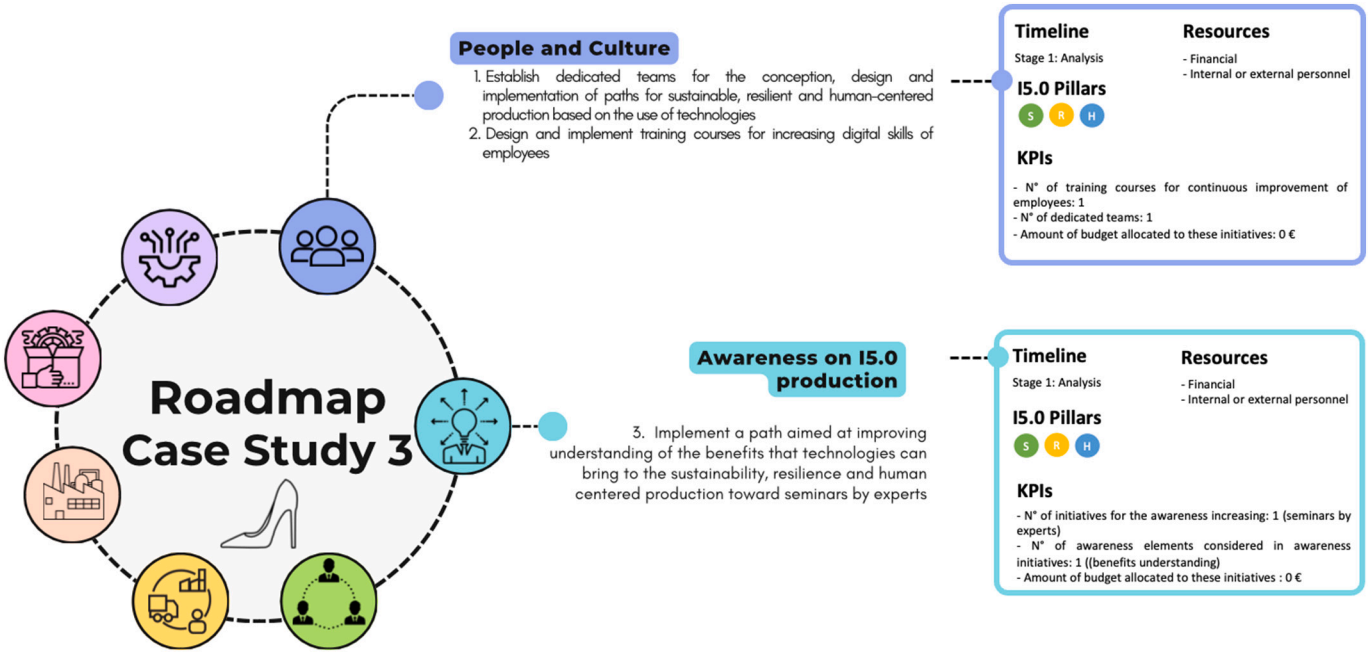


Fig. 14. - Improvement roadmap, case study 3.

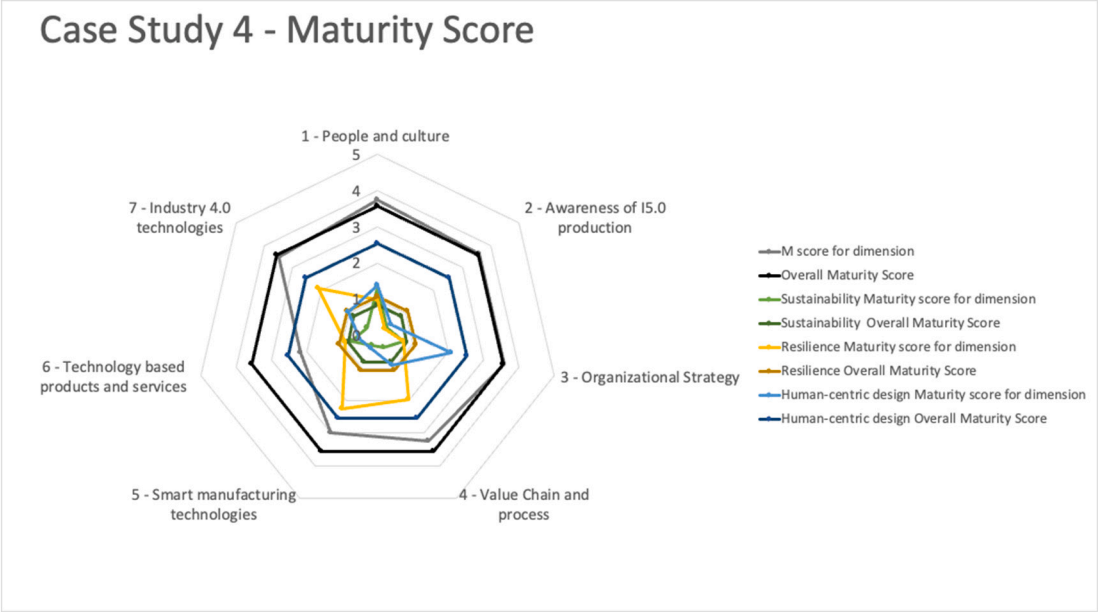


Fig. 15. - Maturity model, case study 4.

geolocating items in case of loss or theft. Other items could incorporate identifiers, such as QR codes printed on inner labels (5.3), which, when scanned, provide handling information (e.g., optimal washing conditions to ensure product durability). These actions are focused on enhancing the customer experience and improving the human-centered design of production while also offering opportunities to increase

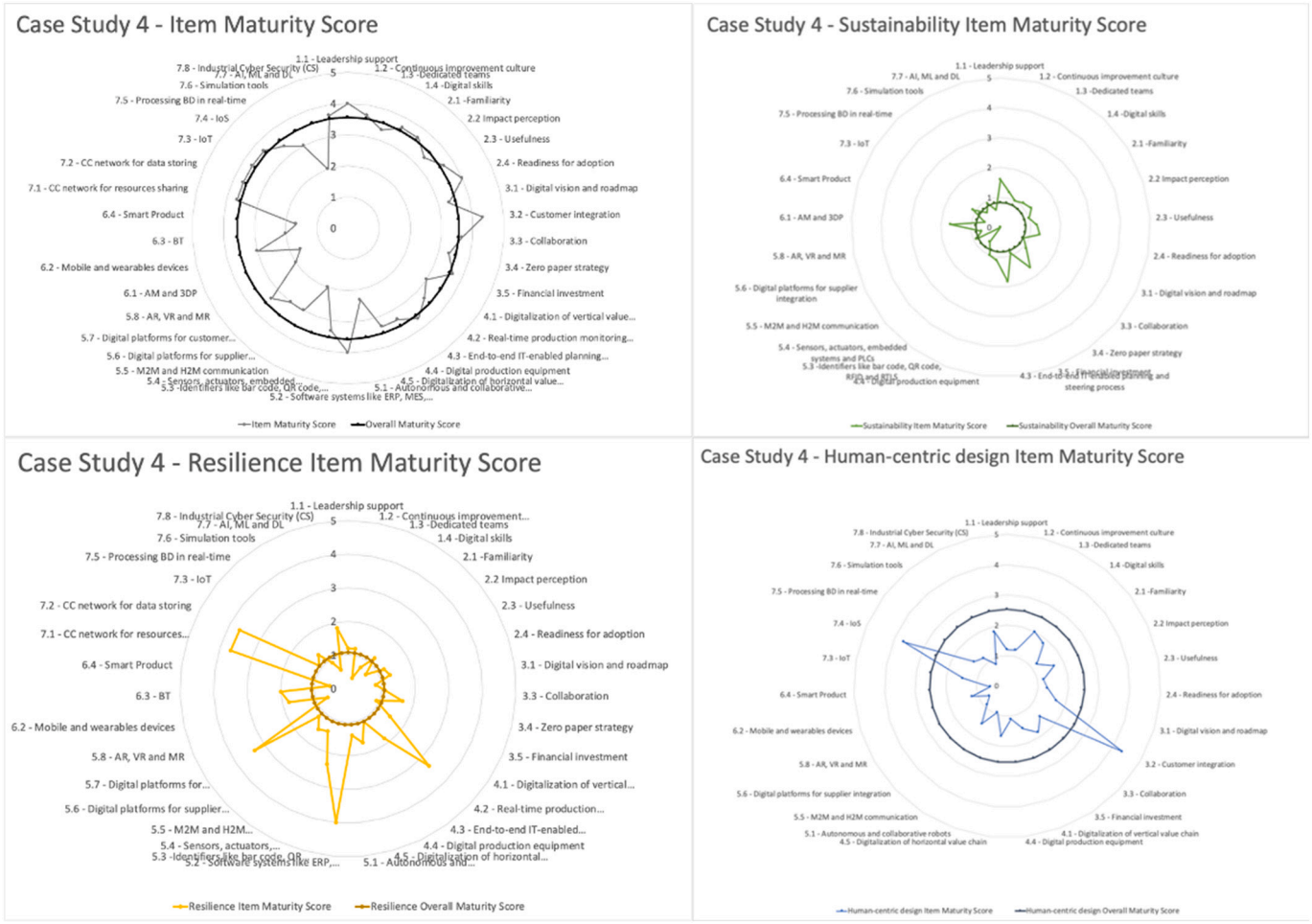


Fig. 16. - Items maturity score, case study 4.

product durability and address sustainability needs. The company is actively engaged in the execution of these actions and has already allocated financial and technological resources to this project. Furthermore, internal personnel (e.g., the production team) is involved in understating how to better integrate sensors into products, working collaboratively with external personnel tasked with technological development.

In addition, the use of augmented or virtual reality tools (5.8) could provide added value. On one hand, these tools could enhance the product design process by supporting customer during customization. On the other hand, they could help the company identify solutions to reduce production waste (e.g., improving the placement of pattern paper on fabrics to minimize waste). For the first case, the company is in the execution phase, as this action is focused on improving its existing product configurator, delivering benefits aligned with the resilience pillar. For the second case, the company is in the design stage, aiming to bring benefits aligned with the sustainability pillar. Notably, financial, personnel and technological resources have already been allocated for these initiatives, which are expected to deliver benefits across all three pillars of I5.0. Fig. 17 summarizes the roadmap for improvement in the Case Study 4.

6. Discussion and implication

The results reported in Section 5 highlight the current attention of SMEs to digital transformation and their awareness of the benefits that I5.0 can bring to sustainability, resilience and human-centered design production. This demonstrates an understanding within the business world of the theoretical models underpinning digital transformation (e.g., I5.0), even though the process still appears to be ongoing.

Three out of four case studies show a good level of experience in leveraging I5.0 technologies to implement sustainable, resilient and human-centered production. However, the maturity level of the various pillars varies across cases. The outcomes of this assessment lead to critical discussions about the areas in which companies needs to focus their improvement efforts. Even when companies demonstrate experience in using digital technologies in production, several areas of improvement can be identified, particularly along specific items and dimensions that reveal weakness. The proposed assessment, conducted as part of a multiple case study, shows that SMEs are aware of the potential of digital technologies to influencing business and society (Ulas, 2019; Verhoef et al., 2021). Additionally, it highlights how smart manufacturing and I4.0 technologies can support this transformation (Rojko, 2017). The findings further reveal that dimensions focusing on organizational aspects are central to a digital transformation journey, confirming the importance of dimensions such as "1 – People and culture" and "2 – Awareness of I5.0 production", as identified by Wagire et al. (2021) and Senna et al. (2023). On the other hand, the overall low maturity levels observed in the use of technologies for products, services and production processes confirms the challenges SMEs face during digitalization (Pelletier and Cloutier, 2019; Wiesner et al., 2018).

It is interesting to note that the two case studies in the agri-food sector recorded the highest maturity score in "2 - Awareness of I5.0 production" and the lowest maturity score in dimension "6 - Technology based products and services". This is likely related to the characteristics of the sector, which has progressed through the stages of digitalization during the industrial revolutions and has adopted Agriculture 4.0 paradigm in farming and production activities (Araújo et al., 2021; Arora et al., 2022; Bento et al., 2019). Moreover, this sector operates closely with nature and people, making companies acutely aware of the environmental and social impacts of production, which generates significant challenges to address (Liu et al., 2021). As result, the sector register high maturity levels in the sustainability and resilience pillars. However, the sector is also characterized by a low-tech product, which, in most cases, can be sold without packaging. This implies that the company can leverage I5.0 through servitization processes, using technology to develop intangible elements that add value to the food product (Kowalkowski et al., 2017). This result opens the door to an interesting area of improvement, enabling agri-food companies to adopt new technologies not only in production processes (e.g., 3D printing in pasta production or high-tech greenhouse for vegetable and fruit production) but also in evolving products toward a servitization process, creating smart agri-food products enriched with an informative layer. By utilizing technologies such as smart label (e.g., sensors, identifiers, IoT) companies can provide final consumer more information about production process and product quality, while also collecting valuable data about consumer habits. The use of web platforms or augmented reality tools can lead to the creation of communities of food enthusiasts who experience the purchasing process in a different and innovative way. These insights represent important feedback that can support the improvement of human-centered product design. Additionally, these technologies can be employed to monitor various stages along the SC, enhancing the resilience of companies in the face of risks or unexpected crises. For instance, during the COVID-19 crisis, the agri-food sector was significantly affected in terms of operations and had to respond by rapidly adapting to new challenges.

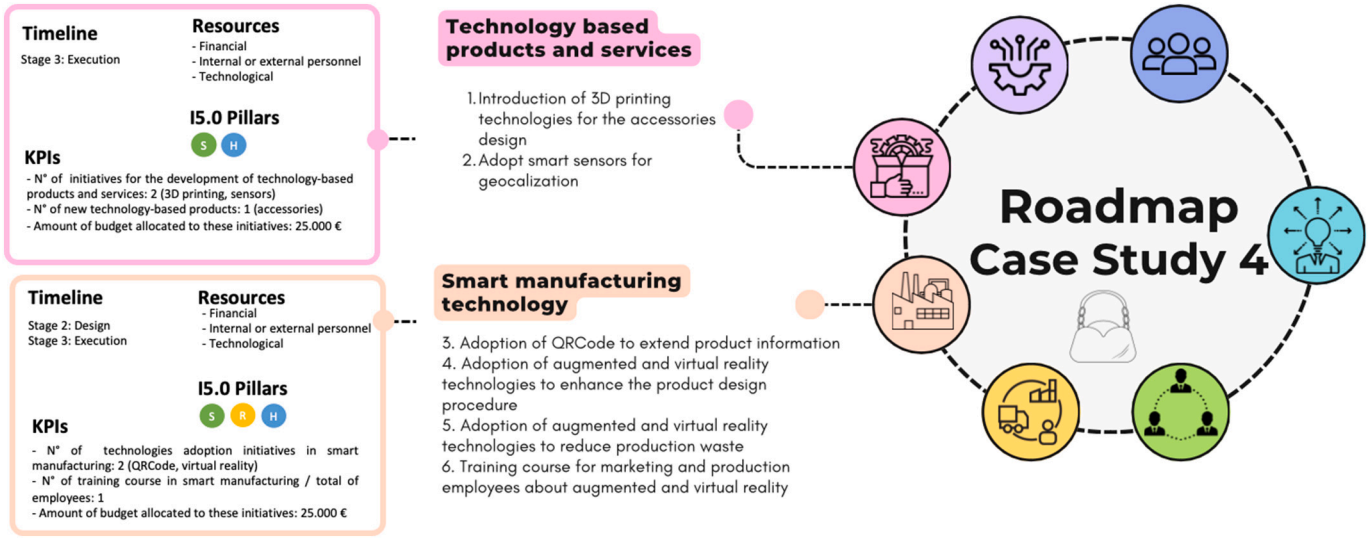


Fig. 17. - Improvement roadmap, case study 4.

The situation described by the case studies of the fashion sector is less standardized compared to the agri-food sector. There remains considerable potential for improvement in the adoption of technologies within production processes, likely tied to the sector's strong artisanal tradition in Italy. It could be valuable to explore how technologies can be introduced into artisanal production processes without altering the image and quality of the products. However, the characteristic of the product (e.g., long-lasting, packaged) make the sector a strong candidate for the development of technology-based products. By incorporating sensors or IoT devices into shoes, bags or clothing, companies could collect data on consumer habits while also providing maintenance and conservation information to extend the product's lifespan, enhancing sustainable and human-centered production.

Thus, the potential to improve the technological level – leveraging technologies that optimize production and, in turn, improve the sustainability of operations (e.g., reducing waste or energy), the resilience of production (e.g., better risk management along the supply chain), and the human-centered design (e.g., integrating consumer into the production process through improved communication) - emerges as a consistent and across need for both industries.

6.1. *Theoretical and practical implications*

The findings of this study reveal several theoretical and practical implications. From a theoretical viewpoint, this study systematically analyzes existing maturity models in I4.0 and I5.0, leading to two important findings:

i) The research field on I4.0 MMs is relatively mature, offering numerous models and empirical results from their application.
ii) The research field on I5.0 MMs is still in its infancy, with no MM focused solely on I5.0 being developed and applied in mature studies published in high-quality journals.

This result identifies a new research gap in MM development, encouraging academia to focus their efforts on creating and testing MMs for the fifth industrial revolution. The proposed I5.0 MM contributes to addressing this gap by focusing on manufacturing SMEs. Furthermore, the mathematical model introduces useful weights to account the importance of dimensions and items within specific application contexts, paving the way for the model's broad applicability in various contexts (e.g., other industries such as healthcare or software, and other types of organizations such as large enterprise or cooperatives). The availability of a ready-to-use questionnaire and roadmap further expands this scope. The robust research methodology, established in alignment with the guidelines of De Bruin et al. (2005), and the experimental application in four case studies, serve as best practice research aiming to develop MM for assessing different aspects of industrial transformation. Finally, the empirical findings of this study reveal that while SMEs are aware of the potential of technologies, they are often unable to fully leverage or implement these technologies effectively across the three I5.0 pillars. This insight can serve as starting point for researchers and practitioners to explore the need for facilitative paths for adopting digital technologies, focusing on overcoming the challenges that companies face (Ramdani et al., 2022)(Mittelstand Digital, 2019; Telekom Deutschland, 2022). Additionally, efforts should be made to clarify the potential of these technologies in achieving sustainability, resilience, and human-centered design. Specifically, researchers and practitioners can address the need for technology integration into agri-food products, providing evidence on servitization paths for such products. Moreover, efforts should focus on showcasing how smart manufacturing or I4.0 technologies can enhance production performance across the three I5.0 pillars and on creating simple tools to measure performance improvements resulting from implementing I5.0 maturity strategies.

From a practical viewpoint, this study offers companies in general, and manufacturing SMEs in particular, the opportunity to reflect on the potential effects of the fifth industrial revolution on all production sectors. These results help companies recognize the importance of moving beyond a technocentric vision of digital development to embrace a more synergistic approach. This approach respects production efficiency while ensuring strong performances in sustainability, resilience and human-centered design.

The study provides significant value by offering a ready-to-use tool that manufacturing SMEs can adopt to self-assess their I5.0 maturity levels. Using the proposed questionnaire and the mathematical model - already equipped with dimensions and items weighted for manufacturing SMEs - a company can evaluate its maturity level, identify areas for improvement across the seven proposed dimensions, and receive feedback on its maturity level in relation to the three I5.0 pillars. Additionally, manufacturing SMEs can benefit from consulting the roadmap for improvement, which offers ideas for establishing appropriate improvement strategies and utilizing KPIs to assess progress.

The empirical findings from the four cases study provide a valuable opportunity for the SMEs in similar sectors (e.g., agri-food and fashion) or other industries to compare their digital transformation progress. The insights gained from these sectors, where digitalization levels remain relatively low, also serve as point of reflection for government bodies and trade associations. These entities can consider implementing incentives to support digital development. For instance, incentives for employee training initiatives could encourage the adoption of I5.0 principles and help SMEs better understand how emerging technologies can enhance the sustainable, resilient, and human-centered aspects of production.

7. Conclusion, limits and follow-ups

This study contributes to the literature in the domain of I5.0 by proposing a tool to assess the maturity of companies in I5.0 practices focusing on sustainability, resilience and human-centred design of manufacturing process. The proposed I5.0 MM is composed by 7 maturity dimensions and 38 maturity items and is applicable using a questionnaire for a self-assessment and a roadmap for improvements. Utilizing FAHP techniques, this study employs a mathematical model that takes into account the dimensions and items weights to reflect the varying importance of dimensions and/or across different application contexts. It was applied in a multiple-case study that involved 4 manufacturing SMEs.

Although the proposed model is supported by prior literature and empirical evidence and several focus groups were established during their development to increase the model validity, there are several limitations and the development procedure still contains a certain level of subjectivity. The proposed questionnaire has the limitation of self-assessment tools, linked to the possible lack of objectivity of the interviewee. Moreover, although the roadmap for implementation was conceived proposing several archetypes of guideline that can support the manager or business analyst in identifying a valid improvement strategy, it is not certain that this list is exhaustive and that a manager or business analyst with little experience will be able to apply it easily. To overcome these limitations, it could be useful to collect empirical results (for example in a large survey) to improve the items and statements of the proposed model and to extend the roadmap over time.

Therefore, the proposed model needs to be tested for its applicability with the aim of generalizing the results in all sectors of manufacturing industries and also in other industries. The application of the proposed I5.0 MM in 4 four case studies from two different manufacturing sectors, emblematic of Made in Italy production, demonstrates that the model can be easily applied to a large sample without requiring specific adjustments for individual companies. Therefore, its transferability to other manufacturing sectors can be immediate and fully advantageous when implementing the last two stages of the adopted methodologies: *distribution and maintenance of the model*. By replicating this analysis

using a broader research methodology (e.g., survey), it would be possible to identify similarities and differences among a large number of companies and trace some typical characteristics of the sector they belong to, understanding how these can be relevant in the adoption of sustainable, resilient and human-centered practices based on the use of technology. It could also be interesting, through the application of stakeholder theory, to explore how each stakeholder acts in the process of adopting I5.0 practices. Finally, considering the possibility of assigning weights to dimensions and items in the model, the MM proposed here, customized to measure the maturity of SMEs in adopting technology-based sustainability practices, could be extended to large companies or cooperative societies through a new weight calculation.

CRediT authorship contribution statement

Maria Elena Latino: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.techfore.2025.124045.

Data availability

Data will be made available on request.

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